

Genetic profiling of the circulating proteome in common diseases suggests causal proteins and improves risk prediction

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In the manuscript, Ma et al presented an interesting work focusing on a large-scale genome-proteome-wide pQTL analysis targeting 2,901 circulating proteins among 7,626 healthy individuals and 28,064 patients across 42 disease types. This study is the first, to my knowledge, to characterize genetic influences on the plasma proteome based on disease-specific status, and further provide insights into shared and specific pQTL between health and disease states. Furthermore, through series of functional annotation and MR analysis, they illustrated the underlying mechanism by which pQTLs influence protein abundance and identified a set of causal proteins contributing to disease risk. Besides, they also constructed a prediction model based on pQTL-PRS and causal ProRS, which can remarkably enhance the accuracy of risk prediction for multiple diseases. Finally, they provide an available database (pQTL-Atlas) for searching pQTLs, discovering causal protein biomarkers and prioritizing innovative drug targets. Overall, this work is well-designed and conducted and technicalities are sound and rigorous, which can be served as a valuable resource to the scientific community and bridge the gap between genome to phenome with plasma proteins, especially in diseases-specific individuals. However, a few issues require further clarification before publication.

Major issues

1. The sample size for this pQTL analysis appears to be somewhat smaller than that of the entire PPP project. Could the authors please elaborate on the reasons for this difference?
2. The authors selected H3K4me1, H3K4me2, H3K4me3, H3K27ac, H3K27me3, and H3K9ac to explore the biological character of pQTL variants. Given the availability of hundreds of histone modifications, it would be helpful if the authors could provide an explanation for choosing these specific histone modifications.
3. Line 192: "28.3% of proteins share a signal with their corresponding gene expression." Why is the number of colocalizing pQTLs appears to be lower compared to previous efforts in the same or other cohorts ($\geq 50\%$)?
4. The survival analysis section is somewhat unclear, particularly regarding the source of the populations used. Could you please provide a more detailed explanation of the screening process for these populations?
5. The authors chose to focus on incident diseases only, omitting prevalent diseases. What is the rationale for omitting proteomic associations with prevalent diseases (i.e., diseases diagnosed at the time of blood draw)?
6. Clinically, the usefulness of risk models is to identify those at high risk of disease in the future, or in some clinical contexts for the diagnosis of disease that is not yet clinically manifest. Since genotype is static, it can make sense to do risk modeling using genotype data on those who already have disease compared to healthy controls. However, protein levels are not static. Did the risk modeling only include those who were disease-free at baseline?
7. While proteomic risk scores may have significant clinical potential, their practical adoption faces challenges similar to the "black box" nature of genomic data. The study does not address this aspect of clinical translation. We suggest acknowledging this limitation and proposing future strategies for feature reduction, possibly focusing on a smaller set of

proteins, such as 20 or 30.

8. The atlas allows public access to the protein-disease association results. Please clarify for readers how they can access the individual level data.

Minor issues

1. Line73: I suggest revising "significantly reduced" to "generally exhibit smaller effect sizes" to clarify that the description refers to absolute effect magnitude rather than statistical significance.

2. Line104: "systematical" should be "systematic".

3. Please explicitly distinguish cis- and trans- pQTLs in the figure or its legend.

4. The term "functional characterization of pQTL" should be changed; while this may be nomenclature in human genetics, "function" implies studies to examine effector function.

5. Line312: Replace "consisted" with "consisting" for grammatical accuracy.

6. The manuscript comprises a vast array of diverse information. The author should enhance the coherence of the text.

7. Please ensure all acronyms are defined at first mention.

Reviewer #2

(Remarks to the Author)

Ma, He, Cai, Li et al. present a pQTL analysis of the UK Biobank Pharma Proteomics Project samples stratified by disease status. They performed pQTL mapping using individuals with 42 different diseases and a healthy subset, identifying millions of disease-specific genetic variants. This represents, to my knowledge, the first large-scale disease-stratified pQTL analysis, making it an intriguing and important analysis to move the field forward. They functionally characterize the pQTLs, classify the extent to which diseases differ in their pQTL landscapes compared to controls, and build predictive models able to discriminate cases from controls with high accuracy. They also detail a pQTL browser that is a welcome addition to make querying the results easier. The manuscript offers an appealing branch into disease-specific analyses, which may better capture disease-relevant genetic regulation than focusing on healthy individuals. However, I have concerns about the interpretation of some of their findings and their methodology that should be addressed.

Major comments

- Throughout the manuscript, the authors focus on the raw number of significant variants instead of on the number of unique associations. While describing the total number of variants associated with protein levels initially is fine, the numbers generally become meaningless. For example, the authors say that 30 associations were identified in lung cancer. Does this represent 30 independent pQTLs, or is this 30 variants, all of which are in the same LD block associated with one protein? I would suggest that instead of focusing on the total number of variants, the authors instead focus on the number of independent associations in all analyses beyond the initial paragraph.
- I'm unclear on how the sample sizes for each disease were derived. While I see the ICD-10 codes listed in ST1, the sample sizes as detailed in ST4 are smaller than I would expect. For example, 132 Parkinson's cases are listed, but when I check the values listed by Sun et al.¹, for just one of the Parkinson's related ICD-10 codes (G20, see ST2 in that manuscript), they list 342 cases in just the randomized baseline portion of the samples. Given both manuscripts utilize the UKB-PPP, why does this discrepancy exist between the studies?
- For the genomic and proteomic QC procedures, it should be stated whether the steps were performed per-disease or in the complete dataset that was then subset to specific diseases.
- Lines 145-147: The extent to which novel associations were identified is surprising, given all of these samples were already included in the full UKB-PPP analysis. It would be beneficial to include a column in ST5 that denotes the novelty of each pQTL.
- Lines 175-177: Do these percentages represent the proportion of the total pQTLs, or the proportion of the independent pQTLs? The latter would be more valuable.
- Line 184: I don't know if this analysis is very informative, as similar analyses have already been performed in larger datasets (e.g. Pietzner et al.² which has a very similar figure). If this section were to be kept, I would suggest focusing on the disease-specific pQTLs instead of on all pQTLs.
- Line 215: Some colocalization results may not be valid, as there is not dense variant coverage of the regions of interest. For example, the MS results in one instance contain only 2 overlapping variants, which is not sufficient to get a reliable colocalization estimate.
- Line 223: The authors reference APOE's relevance to dementia, which is certainly true especially for Alzheimer's disease. However, APOE is not colocalized with dementia risk according to their analysis, nor is it significant through MR. Is this a sample size issue where the APOE pQTL is not found in dementia? If so, this should be discussed.
- Line 228: What background gene set was used to perform the pathway analysis? If the default was used, the results shown are invalid. The background gene set should be matched to those genes that have proteins included in the Olink panel.
- Lines 240-246: Did any of the listed pGenes or other examples have disease-specific pQTLs that reveal potential drug

mechanisms, or were the drugs usually targeting proteins with cross-category pQTLs?

- Line 254: I think more granularity may be needed in the definition of these categories. I would suggest adding a category for disease-biased and health-biased, which would include variants like the SELENOP and MMP7 examples shown in Supp. Fig. 5D. To be disease-specific, I would suggest more stringent criteria than simply “not significant” in health, as in the SELENOP example there is strong evidence of a concordant QTL even if it did not reach the significance threshold in the healthy analysis.
- Line 255: Again, these numbers are not meaningful. How many independent pQTLs are disease- or health-specific?
- Line 268: What does newly-identified mean? The SELENOP pQTL is in high LD ($R^2 = 0.976$) with a pQTL identified in Sun et al (rs230808)1. It also isn't even found as an independent pQTL in Table S5, nor is the MMP7 example. The CHD15 example, however, is very good.
- Figure 4A: How can 89.74% of breast cancer pQTLs be disease-specific and 89.21% be health-specific at the same time?
- Line 287/Figure 4E/F: How was pathway analysis performed? Again, if the background gene set was not set properly, these results may be invalid.
- Line 333: What summary statistics were used to perform the MR analysis? Were only the disease-specific results used?
- Line 337: 189 median disease-protein pairs through MR seems incredibly high. Were only cis variants included in the MR analysis (based on the number of SNPs included going up to 92 according to ST17, I would guess not). If trans-pQTLs were included, MR-Egger and MR-PRESSO are likely not sufficient to account for horizontal pleiotropy. A sensitivity analysis using only cis-pQTLs would be helpful.
- Line 344: Given colocalization between pQTLs and disease GWAS were also performed, how many of those 1,104 also had colocalization support?
- Line 348: For the PLG-MACE association listed, six variants were included in the MR analysis after filtering based on $R^2 < 0.001$. However, of the eight variants associated with PLG in the MACE pQTL analysis (based on information downloaded through the pQTL browser), only two independent LD blocks (based on $R^2 > 0.7$) are seen (rs571848809, rs191555775, rs41272086, and rs11751347 in one block, rs783150, rs12211977, rs1247539, and rs12191307 in the other block) based on LDLink calculations. How can six independent variants be obtained from only two LD blocks? This is based on using the EUR ancestry population reference, which the vast majority of the included pQTL samples should be from.
- Generally, I feel opportunities were missed to dive deeper into similarities and differences across diseases. A few suggestions for things that could be done to investigate this:
 - o Clustering of diseases based on their QTL profiles
 - o Trans-pQTL hotspots that may occur only in certain families of diseases
 - o Cancer-specific pQTLs or pQTLs that only occur in one cancer and not others
 - o Analysis of pQTL directionality and effect size concordance across disease states – do any have discordant effects in different diseases?
- While including a MAF threshold of 0.05 makes sense for smaller studies, it would be interesting to see if more rare variants may be able to uncover additional disease-specific pQTLs when analyzing some of the diseases with larger sample sizes.

Minor Comments

- Line 73: I don't think eQTLs have been shown to have reduced effects on protein levels, and even if they did, this sentence does not say compared to what. Perhaps this should be reworded to say that not all eQTLs also affect protein levels.
- Line 134: When combining all analyses together, how many independent pQTLs are there? i.e. grouping the rs1260326 associations for A1BG (Table S5) together into one pQTL instead of multiple across diseases.
- Line 136: It's not clear where the 47.19% number came from. Neither of the panels in Fig. S2 seem to support that value.
- Line 137: I believe Fig. S2A and S2B are switched.
- Line 155: What settings were used when selecting matched SNPs using vSampler?
- Line 176: It would be interesting to include an example of a pathway-specific pQTL.
- Fig. 3A: It looks like a lot of the pQTLs that have weak eQTLs are from the cancer disease category. Is that the case? Any explanation for why that might be?
- Line 192: What proportion of healthy pQTLs colocalized as a comparison to the 20% here? And what eQTL tissue was used to compare to healthy individuals?
- Line 204: It is interesting to see such substantial differences in GWAS enrichment depending on the pQTL category. Do you have an explanation for why that may be the case?
- Line 209: Did any specific pQTLs contribute substantially to the large pQTL-mediated heritability for any of these traits?
- Line 262: Many of these are likely due to sample size limitations in a number of diseases. For example, DOLM ($n=6867$) vs Breast cancer ($n=778$). Many of the “DOLM-specific” pQTLs would probably be discovered in breast cancer-specific analyses if larger sample sizes were used, so I don't think they should be considered disease-specific. For example, comparing similar sample sizes, only 23% were specific to cataracts vs osteoporosis and only 15.8% were specific to AF vs skin cancer.
- Line 266: These numbers do not seem to match those on line 453 in the discussion.
- Line 323: It would be beneficial to include Kaplan-Meier Curves for a few of these examples for the reader to visualize these trends. Additionally, what is the raw p-value that corresponds to $FDR < 0.05$ for this analysis?
- Line 369: How was the weighted summation of protein levels performed? What were the weights based on?
- Line 465-466: This is not a complete sentence.
- Line 483: The sample size differences between the diseases are an important point and should be discussed.
- Line 700: How were disease-relevant tissues defined?

References

- 1 Sun, B. B. et al. Plasma proteomic associations with genetics and health in the UK Biobank. *Nature* 622, 329-338 (2023). <https://doi.org/10.1038/s41586-023-06592-6>

Reviewer #3

(Remarks to the Author)

In the manuscript "Genetic profiling of circulating proteome for common diseases: Uncovering causal proteins and predicting disease risk.", Ma et al. conducted a comprehensive genome-wide pQTL analysis for over 2,900 plasma proteins across 42 diseases from a cohort of over 35,000 individuals. Authors further performed a number of downstream analyses to explore causal relationships between proteins and diseases, and constructed risk prediction models by integrating pQTL derived polygenic risk scores and causal-protein-derived protein risk scores. Authors also developed an online database for the results of the study, which is a valuable resource. Overall, this is a very comprehensive study with lot of data and analyses and the primary findings of the study seems to be interesting. However, there are several major concerns regarding the research hypothesis and the presentation of the results. The manuscript needs to be more clearly presented with biological or clinical interpretations of the findings. Authors also need to discuss their findings in the light of previous studies. The figures are very dense and hard to interpret. The major concerns are:

1. The main hypothesis of the study is that disease-related pQTLs are present only under certain disease conditions, and thus the pQTL analysis is performed in case-only cohorts for several diseases using the prevalent cases for each disease. Based on the hypothesis, the changes in the protein levels would be the effect or consequences of the disease pathophysiology. However, the authors have further assessed the causality of the proteins using the incident cases and MR analysis. Authors also used the identified pQTLs to predict the diseases. These downstream analyses do not align with the primary hypothesis of the study.
2. How did the authors account for the co-morbidity of the included diseases. What was the proportion of co-occurrence of the diseases in the cohort.
3. There are huge differences in the sample sizes across the diseases (varying from 55 to 12,000 individuals), and hence statistical power for the discovery of pQTLs, which could be one of the potential reasons for differences in the number of identified pQTLs across the diseases. Many of the pQTLs for diseases with large cohort may have not been identified in other diseases due to the smaller sample size (false negative). Could the authors check the significance level of the pQTLs identified in diseases with larger cohort such as hypertension and/or in healthy cohort in the diseases with smaller cohorts. I would suggest having a threshold of minimum number of cases for a disease to be included in the study and restrict the analysis to such diseases.
4. Effect of medication on the pQTL analyses is not discussed.
5. The manuscript especially the results section is very long and difficult to follow with all the numbers and statistics. Result section needs to be presented in a concise and better way.

More specific comments:

Line 127: how are the pQTLs defined in the study? Are they referred to the variants or regions?

Line 129: "Additionally, we found that approximately 50.06% of pQTLs play a role in regulating protein expression in cis and 49.94% of pQTLs function in trans (Fig. 1E)." These are statistical associations and do not necessarily suggest that they play role in regulating protein expression.

Line 142: "To assess the efficacy of our pQTL detection method with the current sample sizes, we compared our detected pQTLs to previously reported pQTLs linked to plasma protein levels. Noticeably, among these 2,339 independent pQTL regions, 1,442 regions (61.65%) have been corroborated by previously published research[5–7], whereas 897 regions (38.35%) were newly identified." Based on this analysis authors concluded that "that existing datasets can capture most known pQTL associations." This can't be concluded from the analysis performed. To do this, authors should check how many of the previously reported pQTLs were also detected in this study, which is not performed here.

Line 153: "we generated a set of controls (non-pQTLs) for comparison, ensuring matching distributions of minor allele frequency to pQTLs." No detail on how this set of non-pQTLs was generated is provided.

Line 173: "we employed a tiered system that classified pQTL associations into protein-specific, pathway-specific and unspecific categories." How were the pQTL associations classified into these categories?

Line 214: Most of the results are presented as average e.g. "colocalization analysis identified an average of six proteins". It would be good to provide range instead of average as the it could be misleading here as the number varies a lot.

Line 228: Term "protein coding genes (pGenes)" has been used at several occasions. It is not mention what does it mean.

Line 237: "Collectively, 98,942 survival-related pQTLs were identified". Are these independent pQTLs? All results should be provided for independent pQTLs to be more meaningful.

Line 392: The C-indices for models with PRS and without PRS are very similar for most of the diseases (Fig 6B), indicating that addition of PRS to the ProRS doesn't add to the prediction. Thus the conclusion that integrating genetic and proteomic data refines predictive accuracy is not correct.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for the careful revisions. I have no further comment.

Reviewer #2

(Remarks to the Author)

I commend the authors for thoroughly addressing the many comments and suggestions from the first round of revisions. I believe the manuscript is greatly improved and the analyses much more complete with the integration of these changes. I have a few additional comments & suggestions to address discrepancies in the manuscript or for potential clarification.

Major:

1. The authors include detail for how individual pQTLs are identified (through conditional analysis) but do not explain how pQTL regions are defined. This should be added to the Methods, especially given the discovery of almost 300 previously undetected pQTL regions. A supplementary table with summary info for each region may also be helpful, which could include region location, # of associated proteins, a list of the proteins, and the novelty of the region. While that information is technically accessible through Table S5, it is difficult to get a clear picture from that table alone which regions are newly identified.
2. Line 234-237 & Fig. 3G: Is it reasonable to suggest that these pathways (like the “Immunoregulatory interactions between a Lymphoid and a non-lymphoid cell” pathway) are contributing to multiple diseases when the pathway is also significantly enriched for proteins with pQTLs in the healthy population? It seems to me that those proteins are simply more likely to have pQTLs regardless of population characteristics, not because of a relationship to disease status.
3. Line 360: The authors give the example of a T2D-derived pQTL that is associated with T2D risk based on MR. Given this pQTL is only significant in people who already have T2D, how could it contribute to risk of developing T2D? Also notable is that AMY2B, based on Cox regression (Table S18), is associated with lower risk of T2D (HR = 0.75), which would seem to go against the MR result.

Minor:

1. There are a number of typos and a few incomplete sentences throughout the manuscript (e.g., line 263-264, line 421-422) that warrant correction.
2. Given Table S4 is referenced when discussing the # of independent/primary pQTLs (line 135), it would be beneficial to add columns to Table S4 for # of primary pQTLs per disease and # of independent pQTLs for disease.
3. Line 302: Describing this pQTL region by only rs17321515 is misleading, as searching that variant in Table S5 returns only a single pQTL instead of six like expected. Perhaps referring to the region ID (e.g., chr19_153) would make it easier for the reader to find details.
4. Line 355: Is the 69.7% number representing the proportion of healthy results that are significant in MR analyses using disease-derived pQTLs? This should be clarified.
5. Line 462: Are the numbers presented here flipped, i.e. it should be 8,380 disease-related and 1,086 disease-specific pQTLs?
6. Line 861 & 866: The references to these tables are no longer correct.
7. Figure 2A: In the enrichment portion of the plot, some cells (i.e., synonymous variants for breast cancer) are significantly enriched for pQTLs, yet the color suggests a $\log_2(\text{OR})$ of 0. While this is likely because there are no synonymous variants in the non-pQTL variants selected (leading to an infinite OR), the presentation is misleading. Perhaps the color should be changed for those examples.
8. Figure 2D: Are these plots properly ordered? In lung cancer, for example, the top plot (labeled cis-pQTL) looks more like the distribution when combining the 2nd and 3rd panels of the plot.

Reviewer #3

(Remarks to the Author)

Authors have adequately addressed all the concerns. I have no more comments.

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Point-by-point responses to reviewer comments

Reviewer #1 (Remarks to the Author):

In the manuscript, Ma et al presented an interesting work focusing on a large-scale genome-proteome-wide pQTL analysis targeting 2,901 circulating proteins among 7,626 healthy individuals and 28,064 patients across 42 disease types. This study is the first, to my knowledge, to characterize genetic influences on the plasma proteome based on disease-specific status, and further provide insights into shared and specific pQTL between health and disease states. Furthermore, through series of functional annotation and MR analysis, they illustrated the underlying mechanism by which pQTLs influence protein abundance and identified a set of causal proteins contributing to disease risk. Besides, they also constructed a prediction model based on pQTL-PRS and causal ProRS, which can remarkably enhance the accuracy of risk prediction for multiple diseases. Finally, they provide an available database (pQTL-Atlas) for searching pQTLs, discovering causal protein biomarkers and prioritizing innovative drug targets. Overall, this work is well-designed and conducted and technicalities are sound and rigorous, which can be served as a valuable resource to the scientific community and bridge the gap between genome to phenome with plasma proteins, especially in diseases-specific individuals. However, a few issues require further clarification before publication.

Response: Thanks for the positive comments. We have addressed the concerns in the following responses.

Major issues

1. The sample size for this pQTL analysis appears to be somewhat smaller than that of the entire PPP project. Could the authors please elaborate on the reasons for this difference?

Response: Thank you for your comment. To test our hypothesis that genetic effects on protein levels may become more pronounced or detectable in specific pathological contexts, we restricted the pQTL analysis to individuals with confirmed disease diagnoses prior to enrolment (prevalent cases), rather than the entire PPP cohort. For the healthy control group, we included participants who showed no recorded incidence of the 42 specified disease types or related medication usage, both at baseline and throughout the six-month follow-up period. This stratification ensured valid comparisons between pathological and healthy populations in the pQTL analysis.

2. The authors selected H3K4me1, H3K4me2, H3K4me3, H3K27ac, H3K27me3, and H3K9ac to explore the biological character of pQTL variants. Given the availability of hundreds of histone modifications, it would be helpful if the authors could provide an explanation for choosing these specific histone modifications.

Response: We selected histone modifications (H3K4me1/2/3, H3K27ac, H3K27me3, H3K9ac) because they represent well-established markers of distinct regulatory elements: H3K4me2/3 and H3K9ac characterise active promoters, H3K4me1 and H3K27ac delineate enhancers, and H3K27me3 denotes repressive domains. These marks were prioritized based on three key reasons: (1) their roles in transcriptional regulation are extensively validated (e.g., via ENCODE), making them mechanistically informative; (2) they exhibit consistent patterns across cell types and datasets, ensuring analytical robustness; and (3) their functional annotations help elucidate how pQTL variants may affect gene regulation and ultimately protein abundance.

3. Line 192: "28.3% of proteins share a signal with their corresponding gene expression." Why is the number of colocalizing pQTLs appears to be lower compared to previous efforts in the same or other cohorts ($\geq 50\%$)?

Response: Thank you for the comment. This difference might be due to the different objective and approach of pQTL-eQTL colocalization analyses. Most notably, our study was designed to characterize genetic regulation of protein levels under disease contexts. And we employed eQTL data from disease-relevant tissues for colocalization analyses, such as TCGA-BRCA for breast cancer and GTEx Pancreas for type 2 diabetes. Gene regulatory mechanisms are often tissue- and pathology-specific. Thus, our strategy enhances the biological relevance and interpretability of the findings, and helps to reveal disease-contextual pQTL-eQTL relationships that may be masked in broader population-level studies. Moreover, our proportion of colocalization between eQTL and pQTL in cancer, as well as the correlation between mRNA and protein levels, are consistent with finding from previous TCGA study, further supporting the biological plausibility of our observations.

4. The survival analysis section is somewhat unclear, particularly regarding the source of the populations used. Could you please provide a more detailed explanation of the screening process for these populations?

Response: Thank you for your comment. Survival analysis cohorts were derived from the UK Biobank using ICD-10 classifications. To ensure data quality and relevance, the following inclusion/exclusion criteria were applied:

- (1) Excluded individuals lacking essential covariates (age, sex, genetic data);
- (2) Excluded individuals who withdrew consent;
- (3) Included individuals whose cause of death was attributed to the specific disease under investigation.

The complete screening methodology is detailed in the **Methods** section ("Identification of survival-associated pQTLs", p. 21, lines 695-706). The final sample size for each disease-specific survival analysis, along with the identified survival-associated variants, is detailed in **Table S13**.

Supplementary table13 : Summary of survival-pQTLs.				
Disease abbreviation	Category	Number of samples	Survival-pQTLs	Survival-pQTL pairs
Breast Ca	Cancer	8,546	39	46
Skin Ca	Cancer	28,173	28	30
Melanoma	Cancer	2,608	9	9
Colorectal Ca	Cancer	4,848	10	10
Prostate Ca	Cancer	9,850	20	20
Lymphoma	Cancer	1,836	5	5
Uterus Ca	Cancer	1,333	4	4
Head and neck Ca	Cancer	1,016	1	1
DOLM	Cardiometabolic	40,559	83	146
MACE	Cardiometabolic	31,240	67	104
AF	Cardiometabolic	22,797	33	34
Hypertension	Cardiometabolic	52,996	152	190
CHD	Cardiometabolic	28,768	81	102
T2D	Cardiometabolic	18,781	52	54
PAD	Cardiometabolic	12,723	26	29
HF	Cardiometabolic	9,766	15	16
VT	Cardiometabolic	5,309	35	42
AAA	Cardiometabolic	3,018	1	1
CS	Cardiometabolic	6,974	3	3
HS	Cardiometabolic	2,255	3	3
GERD	Digestive	35,714	12	15
Liver	Digestive	15,786	32	40
UC	Immune	1,989	1	1
RA	Immune	5,041	16	16
SLE	Immune	238	9	9
Crohn	Immune	1,049	4	5
MS	Immune	447	10	16
Osteoporosis	Musculoskeletal	12,693	17	17
Parkinson	Psychiatric_neurolog	2,266	4	4
Dementia	Psychiatric_neurolog	5,845	1	1
Asthma	Respiratory	11,984	82	113
COPD	Respiratory	16,471	66	69
Renal	Urinary	30,612	48	49

5. The authors chose to focus on incident diseases only, omitting prevalent diseases. What is the rationale for omitting proteomic associations with prevalent diseases (i.e., diseases diagnosed at the time of blood draw)?

Response: Thank you for your comment. Focusing on incident cases ensure proteomic measurements precede disease onset, thereby avoiding reverse causation (where the disease state or its treatment confounds protein profiles). This approach follows the fundamental temporal sequence of cause preceding effect. Consequently, it reduces bias and enables direct estimation of disease risk, strengthening causal inference regarding disease-initiating mechanisms. While potentially reducing sample size, prioritizing incident cases enhances the validity of causal inference for identifying disease-initiating mechanisms.

6. Clinically, the usefulness of risk models is to identify those at high risk of disease in the future, or in some clinical contexts for the diagnosis of disease that is not yet clinically manifest. Since genotype is static, it can make sense to do risk modeling using genotype data on those who already have disease compared to healthy controls. However, protein levels are not static. Did the risk modeling only include those who were disease-free at baseline?

Response: Thank you for your feedback. We fully agree that the clinical value of risk models depends on their ability to identify individuals at high risk of developing future disease. In alignment with this principle and accounting for the dynamic nature of protein levels, our risk modeling was conducted specifically among disease-free individuals at baseline. Prevalent cases were excluded to evaluate the predictive performance of the risk scores for incident disease. This methodology is described in the “Disease-risk prediction with different risk scores” section of the **Methods** (p. 27, lines 896–908).

7. While proteomic risk scores may have significant clinical potential, their practical adoption faces challenges similar to the "black box" nature of genomic data. The study does not address this aspect

of clinical translation. We suggest acknowledging this limitation and proposing future strategies for feature reduction, possibly focusing on a smaller set of proteins, such as 20 or 30.

Response: We thank the reviewer for raising this critical translational consideration. Notably, in our revised manuscript, the number of high-confidence proteins used for disease risk prediction was all less than 30, providing a foundation for clinically feasible implementation. (Fig. 5D, Table S21).

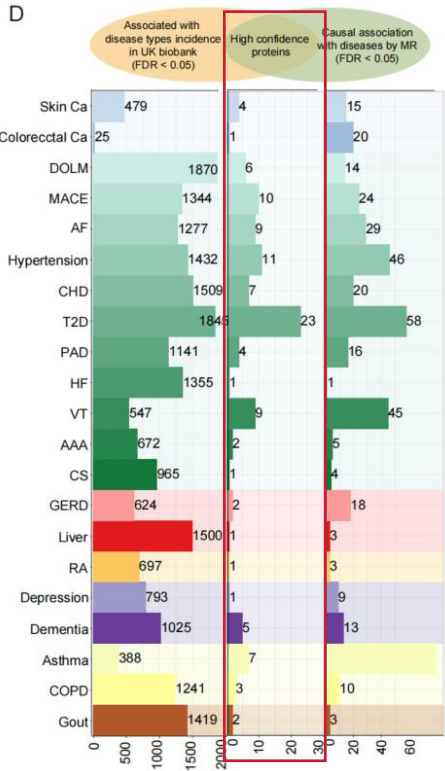


Fig.5D Bar charts illustrate the number of proteins associated with each disease. Left: Protein-disease associations evaluated by cox models; Right: Causal relationship identified by MR. The intersection reveals high-confidence proteins, both achieving FDR < 0.05

In page 11, lines 366-370, we now state:

“To establish robust causal associations between proteins and diseases, we integrated evidence from association analysis and MR. Consequently, 110 disease-protein associations were supported by both analyses and considered highly confident, ranging from 1 for cerebral stroke (CS), colorectal cancer (colorectal ca), Depression, heart failure (HF), liver and rheumatoid arthritis (RA) to 23 for T2D (Fig. 5D and Table S21).”

8. The atlas allows public access to the protein-disease association results. Please clarify for readers how they can access the individual level data.

Response: Thanks for pointing it out, instructions for accessing the individual level data are included in the Data Availability statement (p.31, lines 1053-1056).

Minor issues

1. Line73: I suggest revising "significantly reduced" to "generally exhibit smaller effect sizes" to

clarify that the description refers to absolute effect magnitude rather than statistical significance.

Response: We thank the reviewer for this nuanced clarification. The original statement might be overly absolute and contained inaccuracies. To avoid misinterpretation we have refined the wording accordingly.

In page 4, lines 77-78, we now state:

“However, QTLs affecting mRNA levels are, on average, attenuated or buffered at the protein level.”

2. Line104: “systematical” should be “systematic”.

Response: We are grateful to the reviewer for highlighting this correction. The term has been revised to "systematic" in line 112.

3. Please explicitly distinguish cis- and trans- pQTLs in the figure or its legend.

Response: Thank you for your comment. In **Figure 1B**, darker shades represent cis-pQTLs and lighter shades denote trans-pQTLs, distinguishing the two signal types. This classification criterion is explicitly stated in the figure legend (line 1073) of the revised manuscript.

In page 32, lines 1073-1074, we now state:

“The sample size of each disease type (bottom) and the number of pQTL associations (*cis*, darker; *trans*, lighter) identified for each disease (top).”

4. The term “functional characterization of pQTL” should be changed; while this may be nomenclature in human genetics, “function” implies studies to examine effector function.

Response: Thank you for your comment. We have revised the term "functional characterization of pQTLs" to "Enrichment analyses of pQTLs in regulatory elements" (p.21, line 708) to avoid ambiguity.

5. Line312: Replace "consisted" with "consisting" for grammatical accuracy.

Response: We are grateful to the reviewer for highlighting this correction. The term has been revised to "consisting" in line 322.

6. The manuscript comprises a vast array of diverse information. The author should enhance the coherence of the text.

Response: Thank you for this valuable suggestion. To enhance the manuscript's coherence and logical flow, we have added concise topic sentences at the beginning of each **Results** section. These sentences provide the necessary biological background and state the specific questions addressed in that section. These revisions ensure a clear, step-by-step progression from biological questions through methods and results to interpretation, thereby significantly improving narrative continuity while maintaining scientific rigor.

7. Please ensure all acronyms are defined at first mention.

Response: Thank you for your comment. We have reviewed the manuscript and ensured that all abbreviations are now defined upon first use in the revised manuscript.

Reviewer #2 (Remarks to the Author):

Ma, He, Cai, Li et al. present a pQTL analysis of the UK Biobank Pharma Proteomics Project samples stratified by disease status. They performed pQTL mapping using individuals with 42 different diseases and a healthy subset, identifying millions of disease-specific genetic variants. This represents, to my knowledge, the first large-scale disease-stratified pQTL analysis, making it an intriguing and important analysis to move the field forward. They functionally characterize the pQTLs, classify the extent to which diseases differ in their pQTL landscapes compared to controls, and build predictive models able to discriminate cases from controls with high accuracy. They also detail a pQTL browser that is a welcome addition to make querying the results easier. The manuscript offers an appealing branch into disease-specific analyses, which may better capture disease-relevant genetic regulation than focusing on healthy individuals. However, I have concerns about the interpretation of some of their findings and their methodology that should be addressed.

Response: Thanks for the positive comments. We have addressed the concerns in the following responses.

Major comments

1. Throughout the manuscript, the authors focus on the raw number of significant variants instead of on the number of unique associations. While describing the total number of variants associated with protein levels initially is fine, the numbers generally become meaningless. For example, the authors say that 30 associations were identified in lung cancer. Does this represent 30 independent pQTLs, or is this 30 variants, all of which are in the same LD block associated with one protein? I would suggest that instead of focusing on the total number of variants, the authors instead focus on the number of independent associations in all analyses beyond the initial paragraph.

Response: Thanks for the thoughtful suggestion. As requested, we have performed conditional analyses with GCTA-COJO to define independent signals and focused on these independent pQTLs in the subsequent analyses. In the lung cancer analysis, the 30 pQTL-protein association pairs collapse to 2 independent signals. We have revised the text and figures accordingly to provide a clear characterization of the independent pQTLs.

In page 5, lines 131-134, we now state:

“Furthermore, to discover the independent signals associated with proteins, we performed conditional analyses with GCTA-COJO, yielding a total of 16,562 primary pQTL associations across 2,339 independent genetic regions, spanning from 4 regions in leukemia to 2,221 regions in hypertension (**Fig. 1B, Fig. S2A and Table S5**).”

In page 6, lines 162-167, we now state:

“The results showed that pQTLs accounted for an average of 15.24% of the variance, ranging from 7.37% in hypertension to 70.24% in lung cancer (**Fig. 2D and Table S7**). Interestingly, cis-pQTLs contributed significantly more to the variance (13.50%) compared to trans-pQTLs (1.81%). In addition, we noted that the phenotypic effect sizes of pQTLs were decreased within the increasing of minor allele frequency and cis-pQTLs exhibited a stronger effect in comparison to trans-pQTLs (**Fig. 2E**).”

In page 6, lines 173-177, we now state:

“For each disease type, an average of 83.14% (91.80% in *cis*, 48.87% in *trans*) of pQTL associations were assigned as protein-specific, while 4.58% (4.75% in *cis*, 5.51% in *trans*) as pathway-specific and a further 12.27% (4.45% in *cis*, 40.97% in *trans*) as unspecific (Fig. 2F and Table S8), indicating that pQTLs exert their effects on the circulating proteome through diverse mechanisms.”

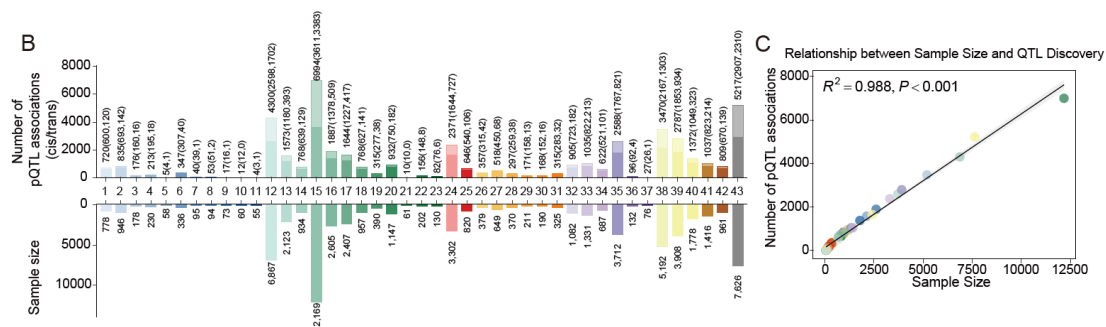


Fig.1B. The sample size of each disease type (bottom) and the number of pQTL associations (*cis*, darker; *trans*, lighter) identified for each (top). The number of *cis* and *trans* are both highlighted in gray.

Fig.1C. The number of pQTL associations against sample size across 43 disease types. The R -squared and P -values of the linear regression model are also displayed.

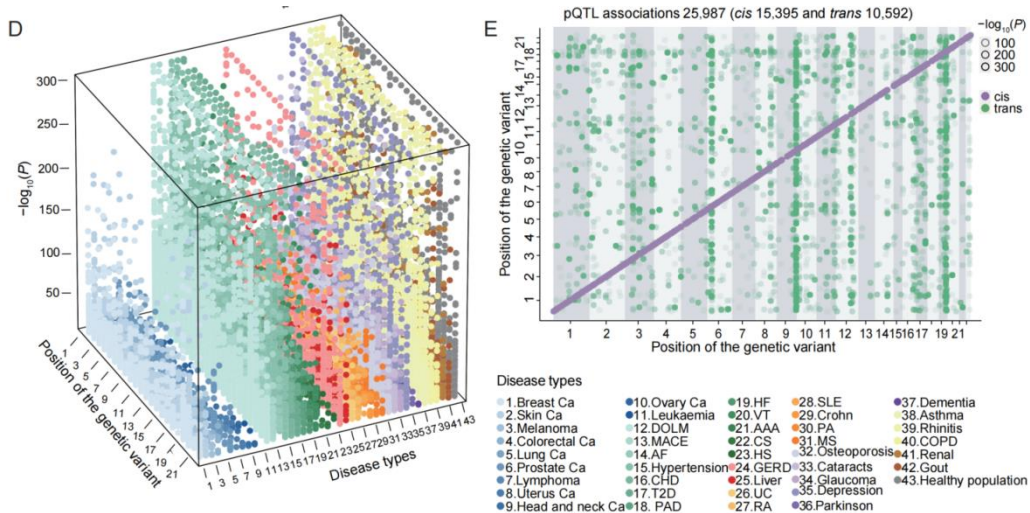


Fig.1D A three-dimensional Manhattan plot displaying chromosomal position (x-axis) of pQTL associations ($P < 1.72 \times 10^{-11}$, z-axis) across 43 disease types (y-axis). Colors indicate disease types.

Fig.1E. Genomic locations of pQTL associations across 43 disease types (*cis*, purple; *trans*, green). The x axis indicates positions of pQTL, and the y axis indicates the gene encoding the associated protein.

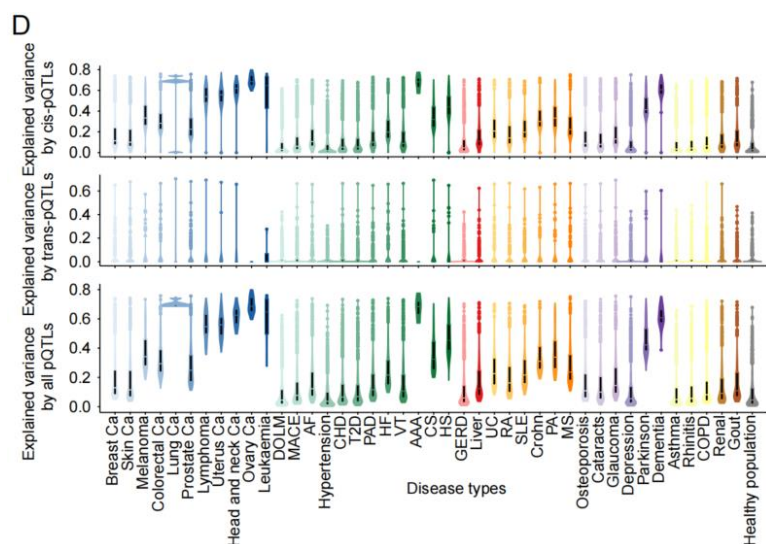


Fig.2D Explained variance in plasma abundances of proteins by identified pQTLs (*cis*, top; *trans*, middle; total, bottom).

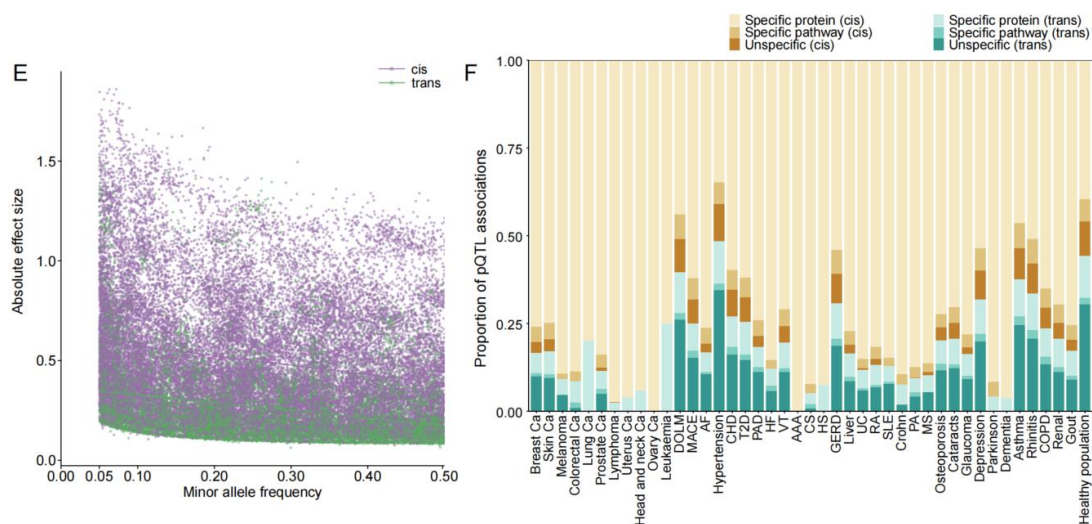


Fig.2E MAF versus effect size of pQTL associations. (*cis*, purple; *trans*, green).

Fig.2F Bar chart showing pQTL classification based on GO term mapping of associated proteins. Yellow shades indicate *cis*-pQTLs and green shades indicate *trans*-pQTLs.

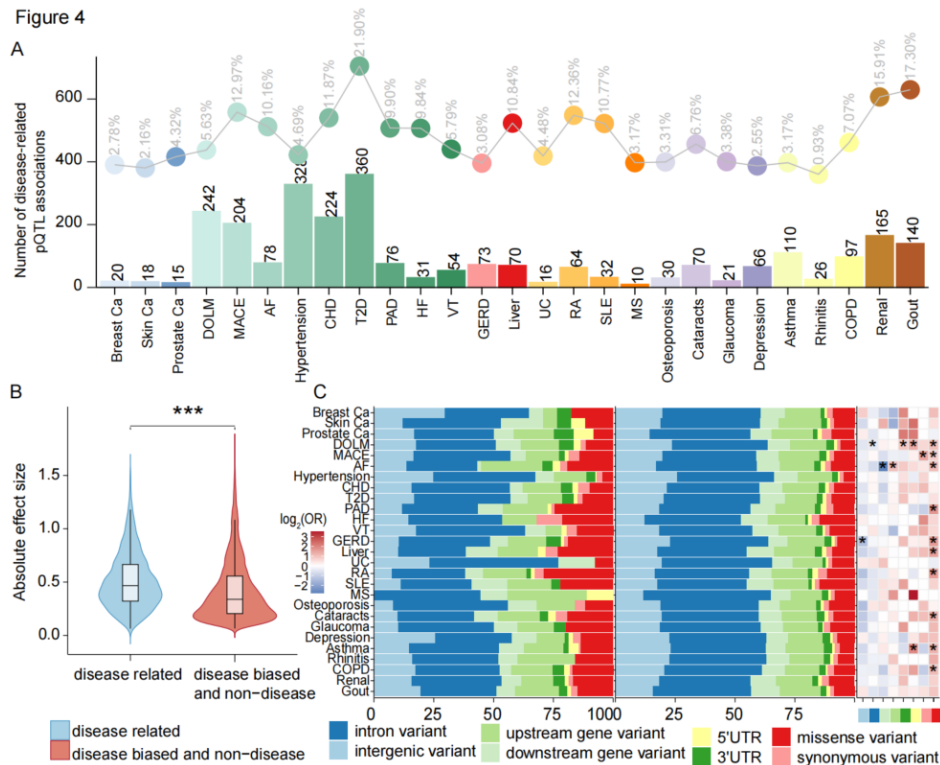


Fig.4A. Bar and Scatter charts show the number and percentage of disease-related pQTL associations

Fig.4B. Comparison of absolute effect sizes of the disease-related pQTL associations (left), disease biased and non-disease associations (right). Two-sided t-test P-value are indicated with "****" if lower than 0.05.

Fig.4C. Genomic distribution of disease-specific (left), disease biased and non-disease (middle) pQTLs, alongside enrichment analysis comparing the two

2. I'm unclear on how the sample sizes for each disease were derived. While I see the ICD-10 codes listed in ST1, the sample sizes as detailed in ST4 are smaller than I would expect. For example, 132 Parkinson's cases are listed, but when I check the values listed by Sun et al.1, for just one of the Parkinson's related ICD-10 codes (G20, see ST2 in that manuscript), they list 342 cases in just the randomized baseline portion of the samples. Given both manuscripts utilize the UKB-PPP, why does this discrepancy exist between the studies?

Response: Thank you for your comment. To test our hypothesis that genetic effects on protein levels may become more pronounced or detectable in specific pathological contexts, we restricted the pQTL analysis to individuals with confirmed disease diagnoses prior to enrolment (prevalent cases), rather than the entire PPP cohort. Additionally, our workflow diagram also highlights this study design.

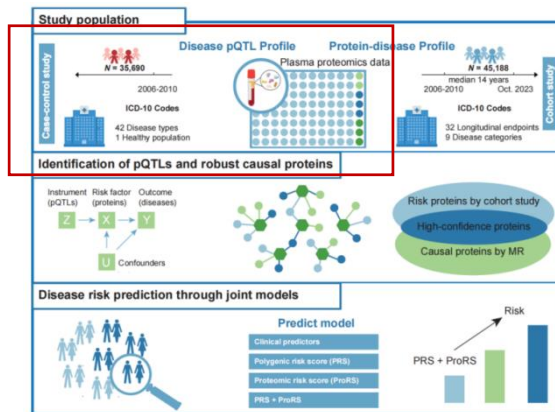


Fig.S10. Abstract of the Study. Firstly, we embarked on a study of the pQTLs in both diseases and healthy population at baseline (at the time of blood sample collection). We extracted data from 35,690 UK Biobank participants, encompassing 42 diseases, a healthy population, plasma proteomics, among other parameters

3. For the genomic and proteomic QC procedures, it should be stated whether the steps were performed per-disease or in the complete dataset that was then subset to specific diseases.

Response: We thank the reviewer for highlighting this important methodological detail. Our quality control (QC) procedures were conducted separately for each of the 42 disease cohorts and the healthy population (43 independent QC workflows in total).

In page 20, lines 656-658, we now state:

“Quality control procedures were executed using PLINK (version 1.90) separately for each disease, resulting in an average of 6,107,244 SNPs per disease type remaining for analysis (**Fig. S1** and **Table S4**).”

In page 20, lines 666-668, we now state:

“Protein quality control was conducted separately for each disease, resulting in the inclusion of 2,895 to 2,902 proteins, with an average of 2,901 proteins analyzed (**Table S4**).”

4. Lines 145-147: The extent to which novel associations were identified is surprising, given all of these samples were already included in the full UKB-PPP analysis. It would be beneficial to include a column in ST5 that denotes the novelty of each pQTL.

Response: Thank you for your insightful comments. As suggested, we have added the new column titled “Type” to denote each pQTL was replicated or newly identified in **Table S5** in the revised manuscript.

5. Lines 175-177: Do these percentages represent the proportion of the total pQTLs, or the proportion of the independent pQTLs? The latter would be more valuable.

Response: Thank you for your insightful comments. These percentages represent the proportion of the total pQTLs. As suggested, we have replaced the proportion of the total pQTLs with that of independent pQTLs. The corresponding results and updates have been incorporated into the revised manuscript (**Fig. 2F** and **Table S8**).

In page 6, lines 173-177, we now state:

“For each disease type, an average of 83.14% (91.80% in *cis*, 48.87% in *trans*) of pQTL associations were assigned as protein-specific, while 4.58% (4.75% in *cis*, 5.51% in *trans*) as pathway-specific and a further 12.27% (4.45% in *cis*, 40.97% in *trans*) as unspecific (Fig. 2F and Table S8), indicating that pQTLs exert their effects on the circulating proteome through diverse mechanisms.”

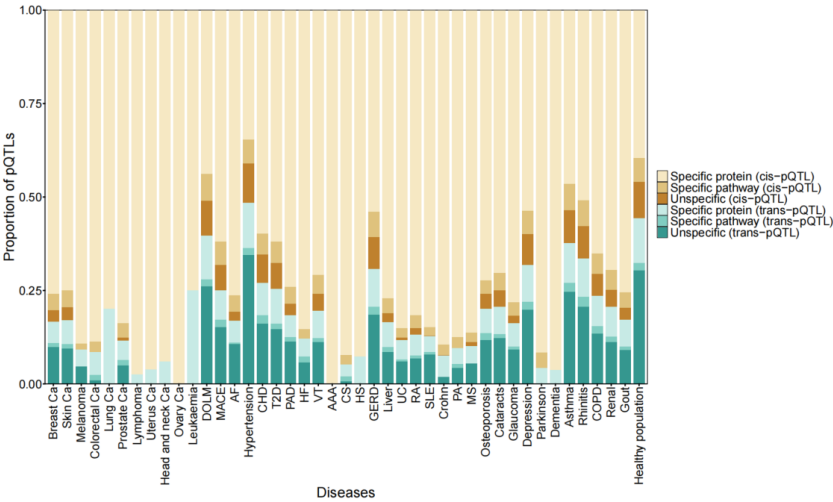


Fig.2F Bar chart showing pQTL classification based on GO term mapping of associated proteins. Yellow shades indicate cis-pQTLs and green shades indicate trans-pQTLs.

6. Line 184: I don’t know if this analysis is very informative, as similar analyses have already been performed in larger datasets (e.g. Pietzner et al.,² which has a very similar figure). If this section were to be kept, I would suggest focusing on the disease-specific pQTLs instead of on all pQTLs.

Response: Thank you for your insightful comment. While we recognize that similar pQTL-eQTL colocalization analyses have been performed in large-scale studies (e.g. Pietzner et al.), our approach differs in several key aspects. Most notably, our study was designed to characterize genetic regulation of protein levels under disease contexts. And we employed eQTL data from disease-relevant tissues for colocalization analyses, such as TCGA-BRCA for breast cancer and GTEx Pancreas for type 2 diabetes.

Gene regulatory mechanisms are often tissue- and pathology-specific. Thus, our strategy enhances the biological relevance and interpretability of the findings, and helps to reveal disease-contextual pQTL-eQTL relationships that may be masked in broader population-level studies. We have revised the corresponding section to better clarify the rationale and distinct focus of our approach.

In page 5, lines 182-187, we now state:

“To elucidate the underlying mechanisms through which pQTLs affect protein abundance under disease contexts and contribute to disease, we undertook a comprehensive evaluation of the shared genetic architecture of protein targets with gene expression, splicing and disease risk.

Given the disease- and tissue-specific nature of gene regulation, we restricted eQTL and sQTL data to disease-relevant tissues, enhancing the biological relevance and interpretability of our

findings.”

7. Line 215: Some colocalization results may not be valid, as there is not dense variant coverage of the regions of interest. For example, the MS results in one instance contain only 2 overlapping variants, which is not sufficient to get a reliable colocalization estimate.

Response: We appreciate the reviewer’s critical assessment. To ensure the reliability of our results, we have updated our analysis to require a minimum of 100 overlapping variants between the pQTL and GWAS datasets for any reported colocalization finding. This threshold provides greater confidence in the colocalization estimates. The corresponding results and updates have been incorporated into the revised manuscript (Fig. 3E and Table S12).

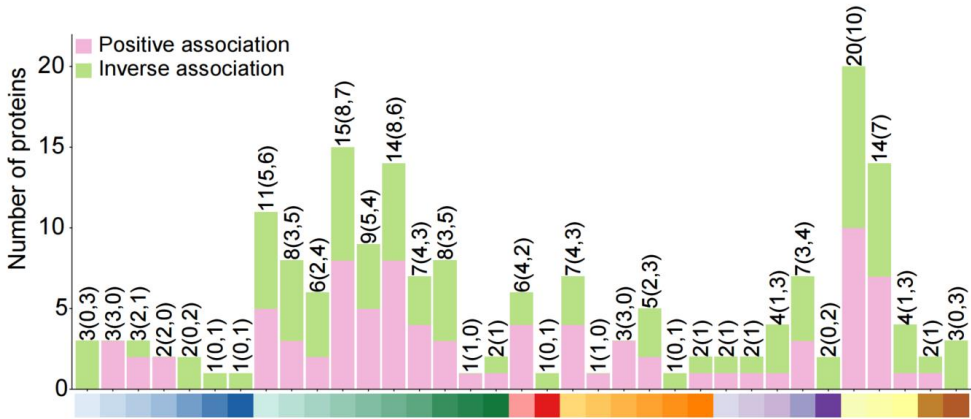


Fig3E. The number of colocalized proteins (posterior probability > 0.7). Colors indicate categories of associations. Pink indicates higher protein abundance and increased risk, while green signifies the opposite.

In page 7, lines 216-220, we now state:

“This identified an average of 5, proteins per disease type sharing genetic determinants (posterior probability > 0.7), ranging from 1 (e.g., uterus cancer (Uterus Ca), ovary cancer (Ovary Ca), abdominal aortic aneurysm (AAA), liver, pernicious anaemia (PA), rheumatoid arthritis (RA)) to 20 (Asthma). These findings demonstrate substantial co-localization between cis-pQTLs and disease loci (Fig. 3E and Table S12).”

8. Line 223: The authors reference APOE’s relevance to dementia, which is certainly true especially for Alzheimer’s disease. However, APOE is not colocalized with dementia risk according to their analysis, nor is it significant through MR. Is this a sample size issue where the APOE pQTL is not found in dementia? If so, this should be discussed.

Response: We thank the reviewer for raising this point. In our pQTL analysis, we identified SNP–APOE associations in the dementia population (min $P = 5 \times 10^{-9}$), which are genome-wide significant at the single-phenotype threshold ($P < 5 \times 10^{-8}$) but do not meet the Bonferroni-corrected threshold ($P < 1.72 \times 10^{-11}$). The current sample size might limit the power to detect associations at this locus. To avoid losing biologically relevant signals, we have expanded Table S12 to include a new “Potential” category for proteins with genome-wide significant pQTLs ($P < 5 \times 10^{-8}$) and designated proteins with pQTLs meeting the multiple-testing-corrected threshold ($P < 1.72 \times 10^{-11}$) as “Confirmed”.

Additionally, we observed strong evidence of colocalization between the APOE pQTL and dementia risk (posterior probability for H4 = 0.93). Thus, we updated Table S12 to include the APOE-dementia colocalization results. Critically, Mendelian Randomization (MR) demonstrated a significant causal effect of elevated APOE protein abundance on increased dementia risk ($\beta = -0.18$, SE = 0.02, FDR = 9.48×10^{-29}). Data for dementia are available in Table S19 and on our website. Taken together, our findings support the pathogenic relevance of APOE protein levels to dementia.

9. Line 228: What background gene set was used to perform the pathway analysis? If the default was used, the results shown are invalid. The background gene set should be matched to those genes that have proteins included in the Olink panel.

Response: Thank you for your insightful suggestions. For each disease, we performed Reactome pathway enrichment analyses. As suggested, we conducted these analyses using a custom background gene set specifically matched to the Olink panel, thereby ensuring statistical validity by accounting for the platform's limited proteomic coverage. The corresponding results and updates have been incorporated into the revised manuscript (**Fig. 3G**).

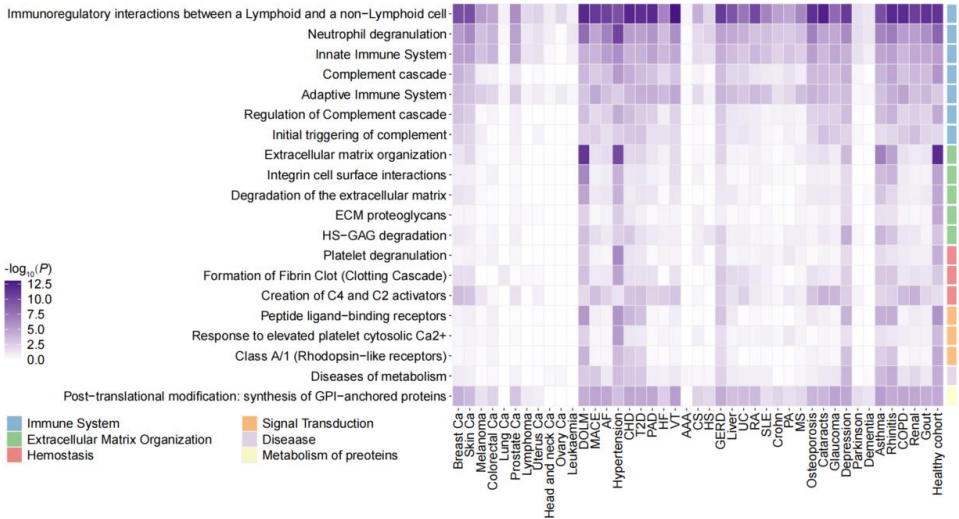


Fig.3G Enrichment results of Reactome pathways for pGenes in each disease type, color-coded by false discovery rate (FDR).

In page 8, lines 231-240, we now state:

“To further elucidate the functional roles of pQTL-regulated protein coding genes (pGenes) in human diseases, we performed pathway analysis on these genes. Among 43 diseases, 34 exhibited significant enrichment of pGenes in at least one Reactome pathway. Immune system-related pathways were the most frequently enriched across diseases (**Fig. 3G**). Specifically, Immunoregulatory interactions between lymphoid and non-lymphoid cells was the predominant immune pathway, observed in nearly all disease types, highlighting lymphocyte-non-lymphocyte interactions may represent a common immunoregulatory hub shared by multiple diseases. Pathways involving protein metabolism and hemostasis, particularly post-translational protein modification: synthesis of GPI-anchored proteins and Creation of C4 and C2 activators, were also enriched in a substantial proportion of diseases suggesting their pivotal involvement in disease pathogenesis.”

In page 24, lines 788-790, we now state:

“For each disease, we performed enrichment analyses for Gene Ontology (GO) biological process terms and Reactome pathways using the pQTL associated proteins, with the set of genes encoding the Olink panel proteins serving as the background.”

10. Lines 240-246: Did any of the listed pGenes or other examples have disease-specific pQTLs that reveal potential drug mechanisms, or were the drugs usually targeting proteins with cross-category pQTLs?

Response: Thank you for your comment. All of the listed pGenes (lines 240-246), including APOE, SMAD3, and VEGFA, have hypertension-related or hypertension-biased (hypertension-specific) pQTLs, supporting their therapeutic potential for hypertension. For instance, the hypertension-biased pQTL rs2965162-APOE ($P_{\text{hypertension}} = 1.09 \times 10^{-19}$; $P_{\text{health}} = 2.04 \times 10^{-10}$), an upstream gene variant, is significantly associated with APOE protein levels specifically in hypertensive samples but not in the healthy controls, indicating stronger regulatory effects under disease conditions. As APOE is critically involved in lipid metabolism and vascular function, this disease-biased regulation offers an underlying mechanistic link to blood pressure modulation, supporting APOE as a promising target for hypertension therapeutics. Similar disease-biased pQTL evidence underpins the potential of targets like SMAD3 and VEGFA.

11. Line 254: I think more granularity may be needed in the definition of these categories. I would suggest adding a category for disease-biased and health-biased, which would include variants like the SELENOP and MMP7 examples shown in Supp. Fig. 5D. To be disease-specific, I would suggest more stringent criteria than simply “not significant” in health, as in the SELENOP example there is strong evidence of a concordant QTL even if it did not reach the significance threshold in the healthy analysis.

Response: Thank you for the thoughtful suggestion. We applied more stringent framework to assess whether genetic effects on protein levels are modified by disease status. As the method mentioned in previous studies (Nathan A, et al. *Nature*. 2022;606(7912):120-128; Haglund A, et al. *Nat Genet*. 2025;57(2):358-368), we re-evaluated each pGenes of the discovered independent pQTLs using linear mixed effects (LME) models in a combined cohort of both disease cases and healthy controls.

We first compared a null model (M0), where protein levels were regressed on age, sex, genetic principal components, batch, and sampling time, against a genotype-including model (M1). Associations where M1 outperformed M0 were considered supported. Across all disease types, all but one association (rs12077761—CGA expression in cataracts) retained significance under this re-evaluation.

We then tested the interaction model (M2) by introducing an interaction term between the disease diagnosis (D) and the genotype (G). A likelihood ratio test (LRT) was used to compare the M2 vs M1 models. Across 27 disease types, we found 2,840 of pQTL associations had a significant genotype-disease interactions (q value < 0.05), indicating genetic effects modulated by disease state. Detailed results are presented in **Fig. 4A** and **Table S14** and are also available on our website.

In page 8, lines 261-268, we now state:

“Next, to further explore how disease status influences the genetic regulation of plasma protein levels, we refitted each independent pQTL association using linear mixed effects (LME) models (**Methods**). We noted the number of disease-related pQTLs correlated with sample size, suggesting expanded. Across 27 diseases with at least 325 cases each, we identified 1,086 disease-associated pQTL associations (q-value <0.05), ranging from 10 in MS to 360 in T2D (**Fig. 4A, Table S14**). We also detected 8,380 disease-biased pQTLs (q-value ≥ 0.05 ; not replicated in healthy cohort) and 14,090 non-disease pQTLs (q-value ≥ 0.05 ; shared with healthy cohort). All results are available on our public database.”

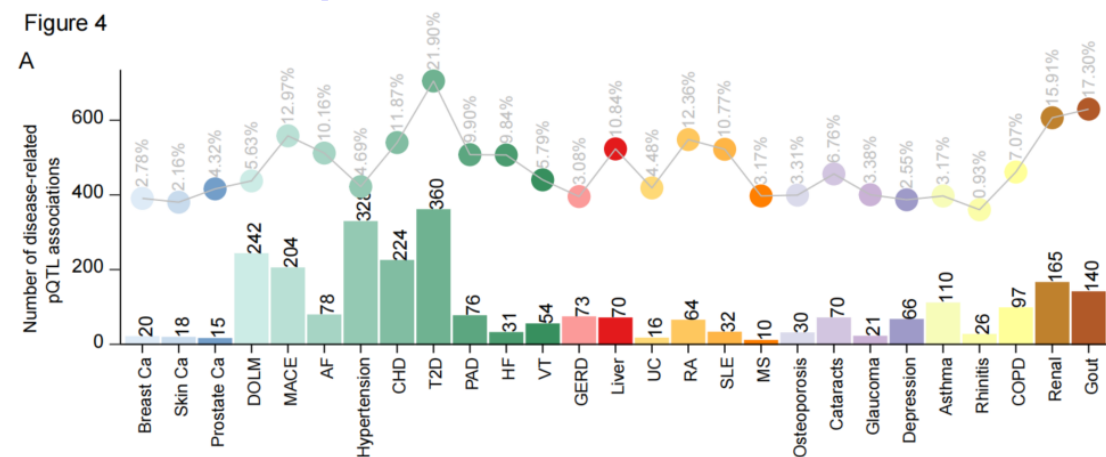


Fig.4A. Bar and Scatter charts show the number and percentage of disease-related pQTL associations

12. Line 255: Again, these numbers are not meaningful. How many independent pQTLs are disease- or health-specific?

Response: Thanks for the helpful suggestion. We detected 1,086 diseases related pQTLs, 8,380 disease-biased pQTLs 14,090 and non-disease pQTLs which also significant in health.

In page 9, lines 264-268, we now state:

Across 27 diseases with at least 325 cases each, we identified 1,086 disease-associated pQTL associations (q-value <0.05), ranging from 10 in MS to 360 in T2D (**Fig. 4A, Table S14**). We also detected 8,380 disease-biased pQTLs (q-value ≥ 0.05 ; not replicated in healthy cohort) and 14,090 non-disease pQTLs (q-value ≥ 0.05 ; shared with healthy cohort). All results are available on our public database.

13. Line 268: What does newly-identified mean? The SELENOP pQTL is in high LD ($R^2 = 0.976$) with a pQTL identified in Sun et al (rs230808)1. It also isn't even found as an independent pQTL in Table S5, nor is the MMP7 example. The CHD15 example, however, is very good.

Response: Thank you for your feedback. We apologize for the inaccuracies in our initial comparative analysis with previous studies, which resulted from the use of improper boundary values. We have now updated the comparison methodology. In line with prior studies, we compared our independent pQTL associations with previously reported lead associations, considering an association replicated if LD $r^2 > 0.1$ (Pietzner M et al. *Science*. 2021;374(6569)). Furthermore, as requested, we now focus specifically on independent pQTL-protein associations

that have not been previously linked to the same plasma proteins (those with $r^2 < 0.1$).

In page 9, lines 273-279, we now state:

“Several newly identified disease-associated pQTL pairs are noteworthy. These include the rs77779567-CDCP1 pair (q-value = 3.92×10^{-4}) linked to major adverse cardiac events (MACE) (Fig. S5C), which is associated with cardiac fibrosis development^[13]; the rs4357365-EPHA4 pair (q-value = 8.30×10^{-3}) associated with breast cancer (Fig. S5D), which is upregulated in the breast cancer stem-like cells^[13]; and the rs4592657-CDH15 (q-value = 1.20×10^{-5}) pair related to coronary heart disease (CHD) (Fig. S5E), which potentially contributes to coronary repeat revascularization.”

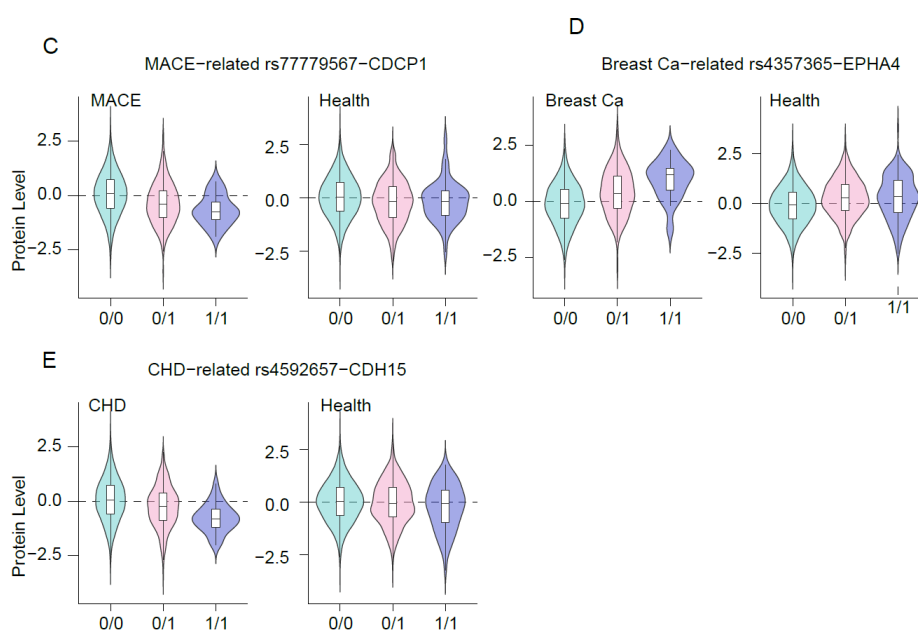


Fig. S5C-E. Examples of disease-related pQTL associations. In the violin plots, the mean and two s.d. are indicated.

14. Figure 4A: How can 89.74% of breast cancer pQTLs be disease-specific and 89.21% be health-specific at the same time?

Response: Thanks for the helpful suggestion. This difference arises because the denominators differ. The denominator for breast cancer cases is the total number of pQTLs identified, whereas for the healthy population it is the number of pQTLs matching those found in breast cancer cases.

15. Line 287/Figure 4E/F: How was pathway analysis performed? Again, if the background gene set was not set properly, these results may be invalid.

Response: Thank you for your insightful suggestions. As suggested, Gene Ontology (GO) biological process (BP) enrichment of pQTL associated proteins was performed using the genes encoding proteins on the Olink panel as the background. The corresponding results and updates have been incorporated into the revised manuscript.

In page 10, lines 304-308, we now state:

“The effect of this hotspot, when acting in a multi-target synergy at this locus in the context of a specific disease, specifically amplifies the co-morbidity pathological network of these diseases in lipid response, inflammation, urea cycle/arginine metabolism, protein stress (**Fig. 4E**), providing crucial clues for understanding their genetic basis and identifying new therapeutic targets.”

In page 10, lines 310-313, we now state:

“The results of pathway enrichment indicate that immune inflammation is the common thread connecting these four seemingly different diseases, and also reflect their respective pathological and physiological characteristics, such as tissue remodeling, barrier damage and repair, and neuroinflammation (**Fig. 4F**).”

In page 21, lines 691-693, we now state:

“For each region, we contrasted our independent results with lead SNPs from previous studies, which were considered replicated when $LD\ r^2 > 0.1$ ^[7].”

In page 24, lines 788-790, we now state:

“For each disease, we performed enrichment analyses for Gene Ontology (GO) biological process terms and Reactome pathways using the pQTL associated proteins, with the set of genes encoding the Olink panel proteins serving as the background.”

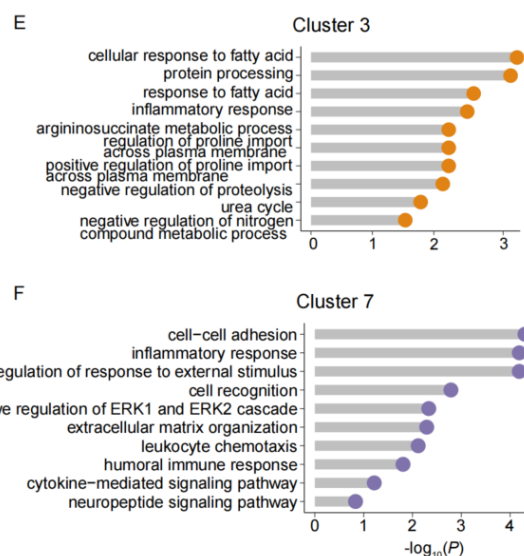


Fig 4E. The results of Gene Ontology (GO) biological process (BP) enrichment proteins associated with the cluster 3-specific trans-pQTL hotspot rs17321515.

Fig 4F. The results of Gene Ontology (GO) biological process (BP) enrichment of proteins related to cluster 7.

16. Line 333: What summary statistics were used to perform the MR analysis? Were only the disease-specific results used?

Response: Thanks for the comment. We conducted two-sample Mendelian randomization. To avoid reverse causation or collider bias due to changes occurring during disease development, we used cis-pQTLs identified in 7,626 healthy individuals as instrumental variables for MR analysis in the revised manuscript. The outcome summary statistics were SNP-disease effect estimates

from GWAS of each disease endpoint, obtained from OpenGWAS and the GWAS catalog.

In page 11, lines 344-346, we now state:

“Subsequently, in order to avoid potential biases introduced by disease-related regulatory changes, such as reverse causation or collider bias, we used cis-pQTLs identified from 7,626 healthy individuals as instrumental variables for Mendelian randomization (MR) analysis.”

17. Line 337: 189 median disease-protein pairs through MR seems incredibly high. Were only cis variants included in the MR analysis (based on the number of SNPs included going up to 92 according to ST17, I would guess not). If trans-pQTLs were included, MR-Egger and MR-PRESSO are likely not sufficient to account for horizontal pleiotropy. A sensitivity analysis using only cis-pQTLs would be helpful.

Response: Thank you for your insightful comment. As described in our Methods, we used single-phenotype genome-wide significant ($P < 5 \times 10^{-8}$) cis-pQTLs as instrumental variables (IVs) for the MR analysis, excluding SNPs within the MHC region. The excessive number of SNPs previously observed likely resulted from the suboptimal distance parameter chosen during clumping, which may have fragmented high-LD regions. To address this, we re-ran the MR analysis in the revised manuscript using a clumping distance of 10 Mb. This adjustment ensures high-LD regions remain intact, aligning with current standard practices in the field (Sun BB et al. *Nature*. 2023;622(7982):329-338; Deng YT et al. *Cell*. 2025;188(1):253-271).

In page 11, lines 344-346, we now state:

“Subsequently, in order to avoid potential biases introduced by disease-related regulatory changes, such as reverse causation or collider bias, we used cis-pQTLs identified from 7,626 healthy individuals as instrumental variables for Mendelian randomization (MR) analysis.”

In page 11, lines 348-353, we now state:

Using the Wald ratio or inverse-variance weighted (IVW) method, 639 significant putative causal associations ($FDR < 0.05$) were identified, with an approximate median of 20 pairs for each disease. (Fig. 5C and Table S19). Genetically predicted elevated levels of 149 proteins were associated with increased disease risk, whereas 166 were protective. Additionally, 64 proteins exhibited opposing directions of association across different diseases.”

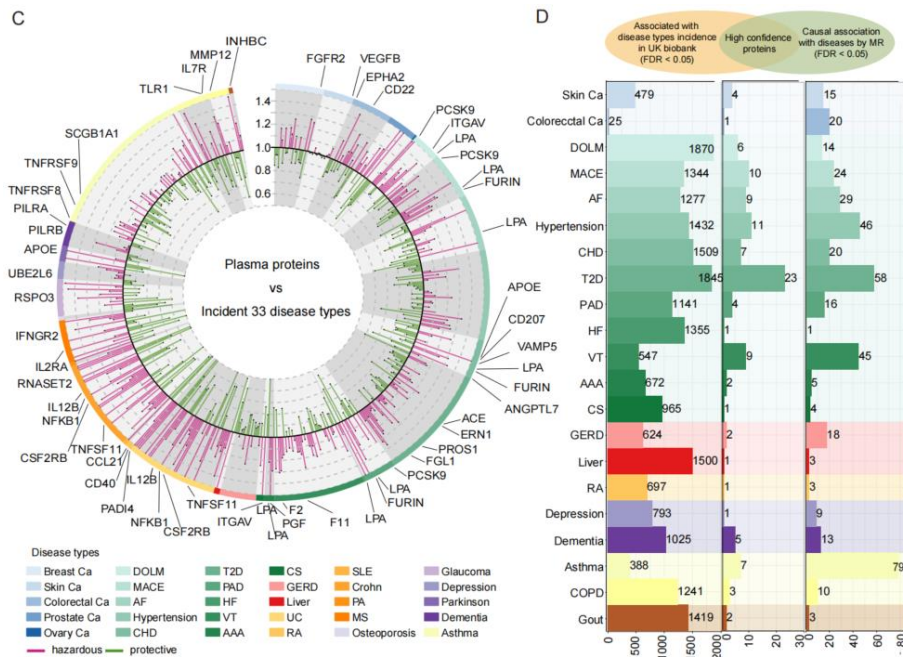


Fig.5C. Causal proteins associated with 33 diseases identified by MR, including multiple proteins that are already targeted with drugs or marked as high-confident (annotated). Each colored block corresponds to a disease type, and each radius represents a protein. The height indicates OR, with red indicating hazardous effects and green indicating protective effects.

Fig.5D. Bar charts illustrate the number of proteins associated with each disease. Left: Protein-disease associations evaluated by cox models; Right: Causal relationship identified by MR. The intersection reveals high-confidence proteins, both achieving FDR < 0.05.

18. Line 344: Given colocalization between pQTLs and disease GWAS were also performed, how many of those 1,104 also had colocalization support?

Response: Thanks for the comment. In the revised manuscript, among the 110 high-confidence protein-disease associations identified, 24 were supported by colocalization analysis.

19. Line 348: For the PLG-MACE association listed, six variants were included in the MR analysis after filtering based on $R^2 < 0.001$. However, of the eight variants associated with PLG in the MACE pQTL analysis (based on information downloaded through the pQTL browser), only two independent LD blocks (based on $R^2 > 0.7$) are seen (rs571848809, rs191555775, rs41272086, and rs11751347 in one block, rs783150, rs12211977, rs1247539, and rs12191307 in the other block) based on LDLink calculations. How can six independent variants be obtained from only two LD blocks? This is based on using the EUR ancestry population reference, which the vast majority of the included pQTL samples should be from.

Response: We appreciate the insightful suggestion. In response to **Comment 17**, we clarify that our initial MR analysis used single-phenotype genome-wide significant cis-pQTLs ($P < 5 \times 10^{-8}$) identified within each disease population as IVs. This differs from the multiple testing-corrected pQTLs ($P < 1.72 \times 10^{-11}$) available on the website. As thus, the number of variants included in the clumping step is therefore significantly larger than eight. Additionally, our initial clumping parameters were suboptimal, potentially truncating high-LD regions and contributing to the

aforementioned issue. To address this, we have re-run the MR analysis in the revised manuscript using a standard clumping distance of 10 Mb. This adjustment ensures high-LD regions remain intact, consistent with current best practices in the field. Applying a stringent LD threshold ($r^2 < 0.001$) within a 10 Mb window retained a single, independent index SNP.

20. Generally, I feel opportunities were missed to dive deeper into similarities and differences across diseases. A few suggestions for things that could be done to investigate this:

Response: We appreciate these constructive suggestions, which provide valuable perspectives for interpreting both shared and disease-specific pathogenic mechanisms based on pQTLs identified across different disease populations. In response, we have expanded our analysis to explore these dimensions more deeply. Specifically, we investigated disease reclassification based on pQTLs profiles and assessed heterogeneity in both the direction and magnitude of pQTL effects among diseases, and performed biological interpretation of the findings. These analyses provide novel insights into the shared and divergent genetic regulatory mechanisms underlying these diseases.

21. Clustering of diseases based on their QTL profiles

Response: Thanks for the constructive suggestions. We applied hierarchical clustering based on magnitudes of pQTL associations (i.e., Betas), and grouped the 27 diseases into 8 clusters.

In page 9-10, lines 293-300, we now state:

“Additionally, we applied hierarchical clustering based on QTL profiles to uncover shared disease mechanisms, and grouped the 27 diseases into 8 clusters (**Fig. 4D, Table S16**), which reveals both well-known connections between diseases and less anticipated similarities. For example, cluster 4 were made up of Ophthalmic diseases including cataracts and glaucoma. (**Fig. 4D**). Common cardiometabolic diseases, disorders of lipoprotein metabolism and other lipidaemias (DOLM), type 2 diabetes (T2D), hypertension and its complications, including renal disorders, also cluster together in terms of their overall pQTL association patterns, as does major adverse cardiac event (MACE) and coronary heart disease (CHD).”

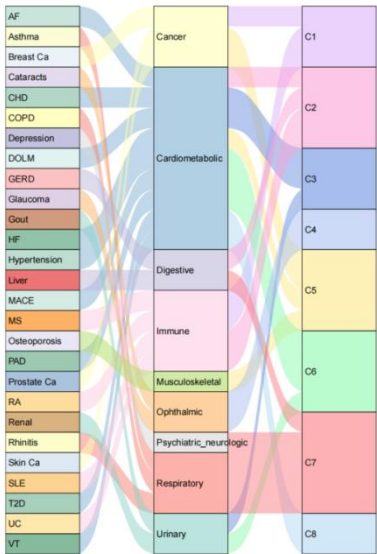


Fig. 4D Sankey plot displays the clustering results of 27 diseases based on pQTL analysis. From left to right, the

figure shows the disease names, their conventional clinical classifications, and the corresponding clustering outcomes.

22. Trans-pQTL hotspots that may occur only in certain families of diseases

Response: We identified a trans-pQTL hotspot occur only in cluster3 including type 2 diabetes (T2D), renal disease (Renal), disorders of lipoprotein metabolism and other lipidaemias (DOLM) and hypertension.

In page 10, lines 300-308, we now state:

“Interesting, there were trans-pQTL hotspots associated with more than ten proteins, occur only in cluster3, underlying the genetic similarity. For example, rs17321515 near 8q24 in TRIB1 was associated with six proteins at $P < 1.72 \times 10^{-11}$ and a further seven at $P < 5 \times 10^{-8}$ (**Fig. S5H**), which is associated with serum lipid levels in humans and cardiovascular risk factors. The effect of this hotspot, when acting in a multi-target synergy at this locus in the context of a specific disease, specifically amplifies the co-morbidity pathological network of these diseases in lipid response, inflammation, urea cycle/arginine metabolism, protein stress (**Fig. 4E**), providing crucial clues for understanding their genetic basis and identifying new therapeutic targets.”

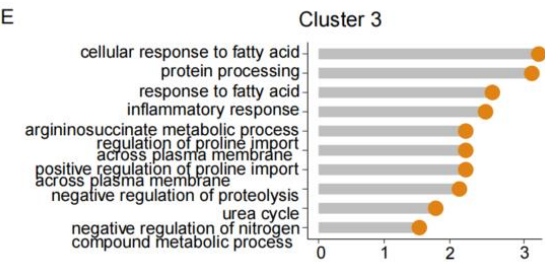


Fig.4E The results of Gene Ontology (GO) biological process (BP) enrichment proteins associated with the cluster 3-specific trans-pQTL hotspot rs17321515.

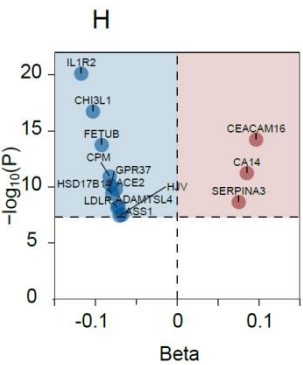


Fig. S5H. The volcano plot displays proteins associated with the trans hotspot rs17321515, specific to cluster 3, showing the lowest p-value across multiple diseases.

23. Cancer-specific pQTLs or pQTLs that only occur in one cancer and not others

Response: Thank you for your comment. We identified three cancer-biased pQTLs that only occur in one cancer and not in other diseases, including colorectal cancer biased rs2005722-PSG1, leukemia-biased rs100536604-GZMH and prostate cancer biased rs113203074-LRPAP1.

24. Analysis of pQTL directionality and effect size concordance across disease states – do any have discordant effects in different diseases?

Response: Thanks for this helpful suggestion. We now highlight pQTLs with significant cross-disease heterogeneity, defined as a greater than twofold difference in effect size or an opposite effect direction. This definition is consistent with previous studies (Li L et al. *Nat Genet.* 2021;53(7):994-1005; Li Q et al. *Nature.* 2022;608(7923):569-577). Overall, we identified 2,832 pQTL associations that have discordant effects in different diseases.

In page 9, lines 282-291, we now state:

“Furthermore, comparative analysis of pQTL effect sizes and directions across diseases may reveal distinct or shared underlying biological mechanisms and potential precision therapeutic targets. In total, we identified 2,832 pQTL associations exhibiting significant cross-disease discordance (twofold or more difference in effect magnitude or opposite direction) across 27 diseases with at least 325 cases each (**Table S15**). For instance, we found 23 pQTL associations with discordance between asthma and rhinitis. These include rs62217915-KIT (**Fig. S5F**), which regulates mast cell proliferation and survival and correlates with asthma severity, and rs8105105-SIGLEC6 (**Fig. S5G**), a dynamically regulated mast cell marker. These associations showed differential effects between individuals with asthma and rhinitis, highlighting genetic disparities and treatment implications between these conditions.”

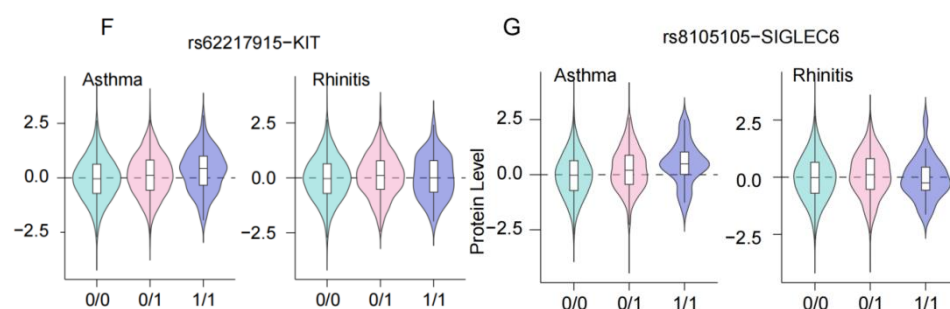


Fig .S5F-G. Examples of disease discordant pQTL associations between asthma and rhinitis. In the violin plots, the mean and two s.d. are indicated.

25. While including a MAF threshold of 0.05 makes sense for smaller studies, it would be interesting to see if more rare variants may be able to uncover additional disease-specific pQTLs when analyzing some of the diseases with larger sample sizes.

Response: Thanks for your insightful suggestion. In this study, we relied on genotype array data, which is optimized for detecting common variants with minor allele frequency (MAF) > 0.05. For the discovery of rare variant associations, whole-exome or whole-genome sequencing approaches would be more suitable. We agree this is an important future direction and plan to incorporate rare variant analyses in future studies with such data.

Minor Comments

1. Line 73: I don't think eQTLs have been shown to have reduced effects on protein levels, and even if they did, this sentence does not say compared to what. Perhaps this should be reworded to say that not all eQTLs also affect protein levels.

Response: Thank you for your comment. The original statement might be overly absolute and contained inaccuracies. To avoid misinterpretation and align with the predominant views in the field (Khan Z et al. *Science*. 2013;342(6162):1100-1104), we have refined the wording accordingly.

In page 4, lines 77-78, we now state:

“However, QTLs affecting mRNA levels are, on average, attenuated or buffered at the protein level.”

2. Line 134: When combining all analyses together, how many independent pQTLs are there? i.e. grouping the rs1260326 associations for A1BG (Table S5) together into one pQTL instead of multiple across diseases.

Response: Thank you for your valuable suggestion. Upon integrating all analyses, we identified a total of 27,454 independent pQTL associations, which include 16,562 primary pQTL associations. We have included these results in the revised manuscript.

In page 5, lines 131-134, we now state:

“Furthermore, to discover the independent signals associated with proteins, we performed conditional analyses with GCTA-COJO, yielding a total of 16,562 primary pQTL associations across 2,339 independent genetic regions, spanning from 4 regions in leukemia to 2,221 regions in hypertension (Fig. 1B, Fig. S2A and Table S5).”

3. Line 136: It’s not clear where the 47.19% number came from. Neither of the panels in Fig. S2 seem to support that value.

Response: Thank you for your careful review and patience. We sincerely apologize for our oversight. The value of 47.19% was calculated based on data underlying Fig. S2B: out of a total of 1,856 pQTL-regulated proteins, 876 were found to be regulated by only one genomic region ($876 / 1,856 \approx 47.19\%$). We have therefore replaced it with a revised version in the updated manuscript to accurately represent these findings.

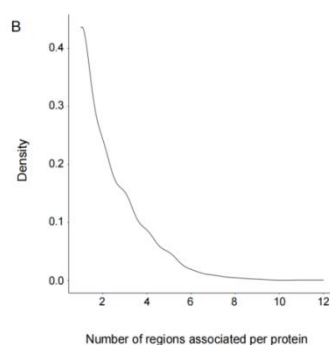


Fig.S2B The number of regions associated per protein.

4. Line 137: I believe Fig. S2A and S2B are switched.

Response: Sorry for the mistake, I apologise once again for our carelessness. The order of Fig. S2A and S2B has been corrected in the revised manuscript.

5. Line 155: What settings were used when selecting matched SNPs using vSampler?

Response: Thanks for your valuable suggestion. The non-pQTLs dataset was randomly sampled by vSampler based on 1000 Genomes Project phase3 (1000G P3), and non-pQTLs were matched with pQTLs based on number of variants in linkage disequilibrium (LD), minor allele frequency (MAF) and variant type (**Fig.S3**). This criterion has been widely adapted in many previous published researches (Takata A et al. *Nat Commun.* 2017;8:14519; Lin D et al. *Genome Med.* 2018;10(1):13; Wang Z et al. *Genomics Proteomics Bioinformatics.* 2023;21(3):649-661) The generation of non-pQTL variants of all disease types was done in the same way.

In page 22, lines 709-713, we now state:

“For comparative analysis, we generated matched non-pQTLs for each disease type by randomly selecting genomic variants using vSampler (<http://www.mulinlab.org/vsampler/>) based on the 1000 Genomes Project Phase 3 (1000G P3). These non-pQTLs were matched to the pQTLs based on the number of linkage disequilibrium (LD)-associated variants, minor allele frequency (MAF), and variant type (**Fig. S3**).”

6. Line 176: It would be interesting to include an example of a pathway-specific pQTL.

Response: Thanks for your valuable suggestion. We have listed a pathway-specific example in our revised manuscript.

In page 6, lines 177-179, we now state:

“For example, the pathway-specific variant rs1761450, associated with LILRB1, LILRB2, LILRB5, LILRA3, and LILRA5 levels in hypertension, coordinately regulates immune receptor activity and cell surface receptor signaling pathways.”

7. Fig. 3A: It looks like a lot of the pQTLs that have weak eQTLs are from the cancer disease category. Is that the case? Any explanation for why that might be?

Response: Thank you for your comment. Protein abundance is influenced not only by transcript levels but also by multiple post-transcriptional and post-translational regulatory processes, including translation rates, translation rate modulation, protein degradation rates, protein synthesis delay and protein transport. Especially during highly dynamic phases, such as cellular differentiation or stress responses, these post-transcriptional processes may lead to greater deviations from the expected correlation between mRNA and protein levels (Liu Y, et al. *Cell.* 2016;165(3):535-550). Moreover, our proportion of colocalization between eQTL and pQTL in cancer, as well as the correlation between mRNA and protein levels, are consistent with finding from previous TCGA study, further supporting the biological plausibility of our observations.

8. Line 192: What proportion of healthy pQTLs colocalized as a comparison to the 20% here? And what eQTL tissue was used to compare to healthy individuals?

Response: Thank you for this important query. Our study was specifically designed to characterize protein regulation in disease contexts, based on the hypothesis that genetic effects on protein levels may be modulated or amplified under pathological conditions. Therefore, we focused our pQTL discovery and colocalisation analyses exclusively on disease-affected

populations. Consequently, no healthy-state pQTLs were generated for colocalisation comparisons.

The observed ~20% colocalisation rate in disease was based on eQTLs derived from disease-relevant tissues (e.g., Pancreas for T2D). In contrast, healthy-state colocalization analysis would require eQTL data from matched tissue of health individuals.

9. Line 204: It is interesting to see such substantial differences in GWAS enrichment depending on the pQTL category. Do you have an explanation for why that may be the case?

Response: Thank you for your comment. We observe stronger GWAS enrichment for blood-related diseases or those with frequent blood exchange (Papier K et al. *Nat Commun.* 2024;15(1):4010), such as heart failure (HF), abdominal aortic aneurysm (AAA), peripheral artery disease (PAD) and renal disease. One explanation is that their disease-related proteins are often secreted or released from tissues into the bloodstream, making them readily detectable in plasma. Consequently, pQTLs for these proteins are more proximal to disease biology, leading to higher GWAS enrichment. Additionally, disease cohorts with smaller sample sizes may also yield less comprehensive pQTL sets, which may weaken the observed enrichment.

10. Line 209: Did any specific pQTLs contribute substantially to the large pQTL-mediated heritability for any of these traits?

Response: Thank you for your comment. Investigating the contribution of disease-specific pQTLs to disease traits is indeed interesting. However, using LDSC to estimate the proportion of disease heritability explained by only a small number of disease-related pQTLs may be challenging (Bulik-Sullivan BK, et al. *Nat Genet.* 2015;47(3):291-5). Notably, our MR analysis revealed some intriguing phenomena. Specifically, we identified proteins exhibiting genetic effects exclusively under pathological conditions, which represent promising therapeutic targets. For example, elevated plasma alpha-amylase 2B (AMY2B) levels were associated with an increased risk of T2D (OR = 1.08, 95% CI = 1.04-1.23). This finding is consistent with the mechanism whereby amylase inhibitors reduce amylase activity, thereby inhibiting starch hydrolysis and lowering blood sugar levels (Chiasson JL et al. *Lancet.* 2002;359(9323):2072-2077).

11. Line 262: Many of these are likely due to sample size limitations in a number of diseases. For example, DOLM (n=6867) vs Breast cancer (n=778). Many of the “DOLM-specific” pQTLs would probably be discovered in breast cancer-specific analyses if larger sample sizes were used, so I don’t think they should be considered disease-specific. For example, comparing similar sample sizes, only 23% were specific to cataracts vs osteoporosis and only 15.8% were specific to AF vs skin cancer.

Response: We agree that disease-specific differences identified previously might partly arise from sample size disparities. In the revised manuscript, we therefore focus on the effect size and direction of pQTLs across diseases, rather than statistical significance alone. Specifically, we highlight pQTLs with significant cross-disease discordance, defined as a greater than twofold difference in effect size or an opposite effect direction. Overall, we identified 2,832 pQTL associations exhibiting discordance in effect sizes and directions across diseases.

12. Line 266: These numbers do not seem to match those on line 453 in the discussion.

Response: Thank you for your comment. We have carefully reviewed the numbers and corrected the inconsistency in the revised manuscript.

13. Line 323: It would be beneficial to include Kaplan-Meier Curves for a few of these examples for the reader to visualize these trends. Additionally, what is the raw p-value that corresponds to FDR < 0.05 for this analysis?

Response: Thank you for your suggestion. In the revised manuscript, we have added the Kaplan-Meier curves as supplementary materials to visually illustrate the survival trends (**Fig.S7**). For statistical significance (FDR < 0.05), the corresponding raw *P*-values ranged from < 0.032 for type 2 diabetes to $< 4.20 \times 10^{-4}$ for colorectal cancer.

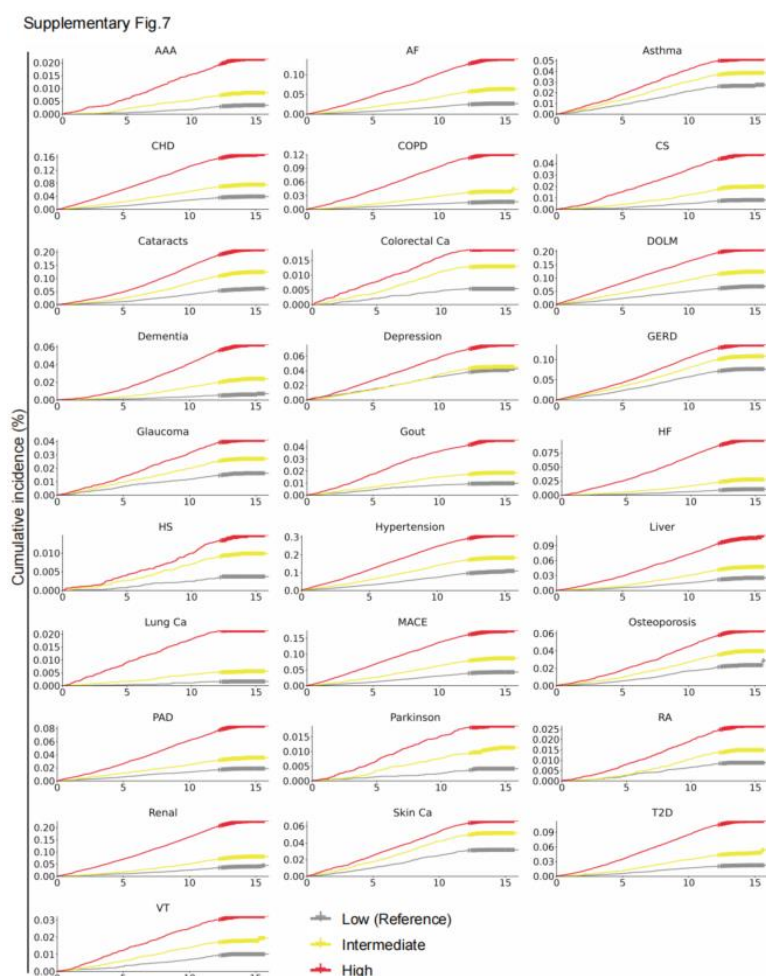


Fig.S7 Kaplan-Meier survival plots of 28 incident diseases according to tertiles of GDF15 levels during follow-up (gray, bottom tertile; yellow, median tertile; red, top tertile).

14. Line 369: How was the weighted summation of protein levels performed? What were the weights based on?

Response: Thank you for your comment. The specific details of how the protein levels were weighted and summed, including the basis for the weights, are described in detail in the **Methods** section titled "Protein Risk Score and Composite Risk Score Development" (p.26, lines 898-911).

15. Line 465-466: This is not a complete sentence.

Response: Thank you for your comment. We have carefully revised the sentence structures to address the noted incompleteness.

16. Line 483: The sample size differences between the diseases are an important point and should be discussed.

Response: Thank you for pointing this out. Our study aimed to provide a comprehensive pQTL resource across a wide range of disease types, and therefore included as many disease types as possible despite variable sample sizes. We agree that differences in sample size across diseases represent an important limitation of our study. For diseases with smaller cohorts, reduced statistical power may have limited the detection of pQTLs and precluded downstream analyses such as causal inference and prediction model validation. We have now expanded the discussion of this limitation in the manuscript and noted that increasing cohort sizes in future studies will be essential to enable more comprehensive and statistically robust analyses across a broader spectrum of diseases.

In page 18, lines 583-587, we added:

“Secondly, for diseases with smaller cohorts, reduced statistical power may limit pQTL detection and preclude downstream analyses such as causal inference and prediction model validation. Therefore, increasing cohort sizes in future studies will be essential for more comprehensive and statistically robust analyses across a broader spectrum of diseases.”

17. Line 700: How were disease-relevant tissues defined?

Response: Thanks, the "disease-relevant tissues" primarily refer to the tissues that are directly affected by or involved in the pathogenesis of the disease. For example, the relevant tissue for cancer is its tissue of origin. In diabetes, it is pancreatic tissue; in asthma and COPD, lung tissue. For systemic cardiovascular and metabolic disorders such as hypertension and dyslipidemia, circulating plasma is considered the relevant tissue.

References

- 1 Sun, B. B. et al. Plasma proteomic associations with genetics and health in the UK Biobank. *Nature* 622, 329-338 (2023). <https://doi.org/10.1038/s41586-023-06592-6>
- 2 Pietzner, M. et al. Mapping the proteo-genomic convergence of human diseases. *Science* 374 (2021). <https://doi.org/10.1126/science.abj1541>

Reviewer #3 (Remarks to the Author):

In the manuscript “Genetic profiling of circulating proteome for common diseases: Uncovering causal proteins and predicting disease risk.”, Ma et al. conducted a comprehensive genome-wide pQTL analysis for over 2,900 plasma proteins across 42 diseases from a cohort of over 35,000 individuals. Authors further performed a number of downstream analyses to explore causal relationships between proteins and diseases, and constructed risk prediction models by integrating pQTL derived polygenic risk scores and causal-protein-derived protein risk scores. Authors also developed an online database for the results of the study, which is a valuable resource. Overall, this is a very comprehensive study with lot of data and analyses and the primary findings of the study seems to be interesting. However, there are several major concerns regarding the research hypothesis and the presentation of the results. The manuscript needs to be more clearly presented with biological or clinical interpretations of the findings. Authors also need to discuss their findings in the light of previous studies. The figures are very dense and hard to interpret. The major concerns are:

Response: Thanks for the positive comments. We have addressed the concerns in the following responses.

1. The main hypothesis of the study is that disease-related pQTLs are present only under certain disease conditions, and thus the pQTL analysis is performed in case-only cohorts for several diseases using the prevalent cases for each disease. Based on the hypothesis, the changes in the protein levels would be the effect or consequences of the disease pathophysiology. However, the authors have further assessed the causality of the proteins using the incident cases and MR analysis. Authors also used the identified pQTLs to predict the diseases. These downstream analyses do not align with the primary hypothesis of the study.

Response: Thank you for raising this important point. To clarify, our primary hypothesis is not that the genetic variants arise only in disease states, but rather that certain genetic variants may exert regulatory effects on protein expression in a disease-state-dependent manner. That is, while the genetic variants are constitutively present prior to disease onset, their functional effects may become detectable or biologically relevant only under specific pathological or stimulating conditions (Farh KK, et al. *Nature*. 2015 ;518(7539):337-343). This hypothesis is supported by evidence showing that the accessibility of certain chromatin regions (Calderon D, et al. *Nat Genet*. 2019;51(10):1494-1505) or the presence of specific epigenetic modifications (Frye M, et al. *Science*. 2018;361(6409):1346-1349) occurs only in disease-specific epigenomic states, thereby enabling the genetic variant to affect translation only in that context.

Based on these mechanisms, using pQTLs identified from disease individuals captures not only the regulatory effects that persist from health to disease but also those that become detectable or amplified only under pathological conditions. As such, they are particularly valuable for polygenic risk score (PRS) construction aimed at improving disease prediction, stratification, or understanding of disease-specific molecular mechanisms.

In order to avoid potential biases introduced by disease-related regulatory changes, such as reverse causation or collider bias, we agree that pQTLs identified from healthy individuals should be used

as instrumental variables for Mendelian randomization (MR) analysis. In response to this comment, we have revised our MR analysis accordingly, and the updated results are presented in **Figure 5C**.

In page 11, lines 344-346, we now state:

“Subsequently, in order to avoid potential biases introduced by disease-related regulatory changes, such as reverse causation or collider bias, we used cis-pQTLs identified from 7,626 healthy individuals as instrumental variables for Mendelian randomization (MR) analysis.

In page 11, lines 348-353, we now state:

Using the Wald ratio or inverse-variance weighted (IVW) method, 639 significant putative causal associations (FDR < 0.05) were identified, with an approximate median of 20 pairs for each disease. (**Fig. 5C** and **Table S19**). Genetically predicted elevated levels of 149 proteins were associated with increased disease risk, whereas 166 were protective. Additionally, 64 proteins exhibited opposing directions of association across different diseases.”

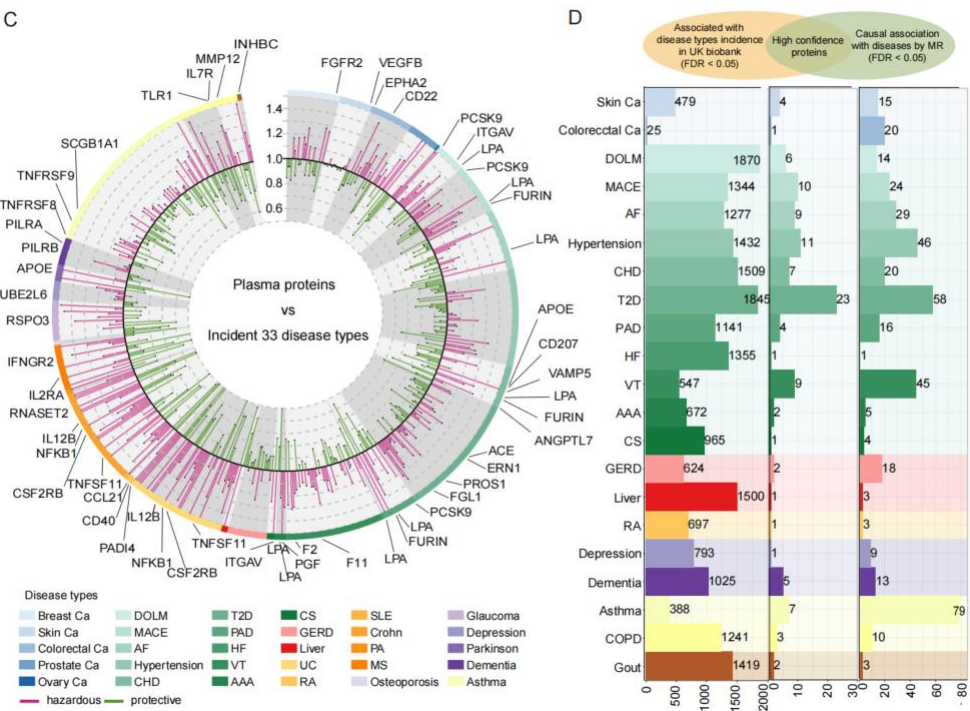


Fig.5C. Causal proteins associated with 33 diseases identified by MR, including multiple proteins that are already targeted with drugs or marked as high-confident (annotated). Each colored block corresponds to a disease type, and each radius represents a protein. The height indicates OR, with red indicating hazardous effects and green indicating protective effects.

Fig.5D. Bar charts illustrate the number of proteins associated with each disease. Left: Protein-disease associations evaluated by cox models; Right: Causal relationship identified by MR. The intersection reveals high-confidence proteins, both achieving FDR < 0.05.

2. How did the authors account for the co-morbidity of the included diseases. What was the proportion of co-occurrence of the diseases in the cohort.

Response: We thank the reviewer for raising this important point regarding comorbidity.

Importantly, strict exclusion of all comorbid individuals would reduce statistical power and may result in a non-representative sample. Therefore, we adopted commonly used strategies from pan-phenotypic studies (Buerger T, et al. *Nat Med.* 2022;28(11):2309-2320) to reduce potential confounding due to comorbidity:

(1) For diseases with established causal links to comorbidities (e.g., cancers), only the first incident diagnosis per cancer type was retained. This reduces the inclusion of secondary or metastatic tumors originating from other primary sites.

(2) For diseases with uncertain relationships to comorbidities (e.g., cardiovascular endpoints), we excluded patients receiving lipid-lowering therapy to reduce confounding.

Nonetheless, we acknowledge that comorbidity remains a general challenge. To further evaluate its potential impact, we conducted a sensitivity analysis in the causal protein-disease association stage. These sensitivity analyses yielded results consistent with our primary findings, suggesting that comorbidity did not substantially affect the observed associations, which is consistent with previous study (Deng YT et al. *Cell.* 2025;188(1):253-271.e7). We have added a description of this analysis in the **Results** section and included the detailed results in **Table S18**.

In page 10, lines 340-342, we now state:

“Similarly, 72.3% of protein-disease associations remained significant when restricting the controls and further adjusting for comorbidity status for each incident disease endpoint (Methods, **Table S18**).”

In page 24-25, lines 825-830, we now state:

“To further assess the influence of comorbidity, we performed a sensitivity analysis with stringent quality controls. Specifically, for each incident disease, we excluded individuals diagnosed with any condition from the same disease category prior to baseline. We also removed participants from the control group who developed an incident disease within the same category. Additionally, we adjusted for baseline multimorbidity from all other disease categories in the Cox models.”

3. There are huge differences in the sample sizes across the diseases (varying from 55 to 12,000 individuals), and hence statistical power for the discovery of pQTLs, which could be one of the potential reasons for differences in the number of identified pQTLs across the diseases. Many of the pQTLs for diseases with large cohort may have not been identified in other diseases due to the smaller sample size (false negative). Could the authors check the significance level of the pQTLs identified in diseases with larger cohort such as hypertension and/or in healthy cohort in the diseases with smaller cohorts. I would suggest having a threshold of minimum number of cases for a disease to be included in the study and restrict the analysis to such diseases.

Response: Thank you for raising this critical issue. The heterogeneity in sample sizes across diseases inherently constrains cross-cohort comparability. In statistical analysis, the *P*-value is particularly sensitive to sample size. We applied the stringent multiple testing correction to limit the power to detect pQTL in smaller cohorts.

As requested by reviewers, we performed sensitivity analyses in hypertension and healthy cohorts

by subsampling to match sample sizes typical of other diseases. We then assessed the repetition rate of previously identified cis pQTL associations (from the full hypertension/healthy cohorts) within these reduced datasets. Our sensitivity analyses revealed that when the sample size falls below 325 samples in healthy cohorts and hypertension cohorts, pQTL detection exhibits pronounced sample size dependency, with curve flattening beyond these thresholds suggesting diminishing marginal returns (**Fig. R1**).

Based on these findings, in investigating similarities and differences in their regulatory effects across various diseases and health states analyses, we included diseases with at least 325 cases, aligning with the healthy and hypertension cohort size threshold.

Finally, to ensure the comprehensiveness of disease representation, we retained small disease cohorts (e.g., cancer, where low incidence results in limited case numbers within population-based cohorts), whose sample size are consist with previous QTL analyses in cancer studies (Li L et al. *Nat Genet.* 2021;53(7):994-1005; GTEx Consortium. *Science.* 2020;369(6509):1318-1330). Meanwhile, we explicitly acknowledge this limitation in the revised Discussion and will verify findings in future, larger datasets.

In page 18, lines 583-587, we now state:

“Secondly, for diseases with smaller cohorts, reduced statistical power may limit pQTL detection and preclude downstream analyses such as causal inference and prediction model validation. Therefore, increasing cohort sizes in future studies will be essential for more comprehensive and statistically robust analyses across a broader spectrum of diseases.”

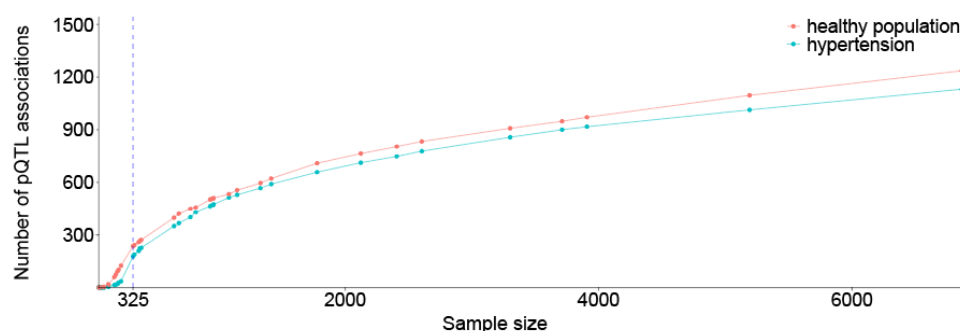


Fig.R1 The scatter plot illustrates the relationship between the subsampling cohort size and the number of replicated cis-pQTL associations.

4. Effect of medication on the pQTL analyses is not discussed.

Response: We thank the reviewer for highlighting the important consideration of medication use in pQTL analyses. In our study, we did not adjust for prescribed medications for the following reasons. First, the UK Biobank dataset lacks precise information on the timing of medication intake relative to blood sample collection, making it difficult to reliably assess the short-term impact of drug exposure on protein levels. Second, protein levels may be influenced by both acute and chronic drug effects, which are highly variable across individuals and drug classes. Given these limitations, we followed the approach of previous studies (Chen et al. *Nat Genet*

2023;55:44-53; Lotta et al. *Nat Genet* 2021;53:54-64) and did not adjust for this factor. We acknowledge this limitation in the **Discussion** section of our revised manuscript.

In page 18, lines 589-591, we now state:

“Lastly, we did not adjust for medication use in pQTL analyses. This factor should be considered in future studies when precise medication timing and protein measurement data become available in UK Biobank.”

5. The manuscript especially the results section is very long and difficult to follow with all the numbers and statistics. Result section needs to be presented in a concise and better way.

Response: Thank you for this suggestion. In the revised manuscript, we have streamlined the content of the main figures to improve clarity. Specifically, in Figure 4, we retained only the core analytical components to reduce complexity and enhance readability. These adjustments were made to present the Results section more concisely and to address the concerns regarding its length and accessibility.

More specific comments:

1. Line 127: how are the pQTLs defined in the study? Are they referred to the variants or regions?

Response: Thank you for your comment. In this study, a pQTL was identified when a SNP demonstrated genome-wide significant association with plasma protein abundance ($P < 1.72^{-11} = 5 \times 10^{-8}/2901$), which refers to specific variant-level associations (SNPs).

2. Line 129: “Additionally, we found that approximately 50.06% of pQTLs play a role in regulating protein expression in cis and 49.94% of pQTLs function in trans (Fig. 1E).” These are statistical associations and do not necessarily suggest that they play role in regulating protein expression.

Response: Thank you for highlighting this distinction. We confirm these findings represent statistical associations between genetic variants and protein abundance, not necessarily functional regulation. The manuscript has been revised accordingly.

In page 5, lines 137-138, we now state:

“Additionally, approximately 59.24% of pQTLs were local acting (‘cis’ pQTLs) and 40.76% were distant acting (‘trans’) (Fig. 1E).”

3. Line 142: “To assess the efficacy of our pQTL detection method with the current sample sizes, we compared our detected pQTLs to previously reported pQTLs linked to plasma protein levels. Noticeably, among these 2,339 independent pQTL regions, 1,442 regions (61.65%) have been corroborated by previously published research[5–7], whereas 897 regions (38.35%) were newly identified.” Based on this analysis authors concluded that “that existing datasets can capture most known pQTL associations.” This can’t be concluded from the analysis performed. To do this, authors should check how many of the previously reported pQTLs were also detected in this study, which is not performed here.

Response: Thank you for pointing this out. We have revised the corresponding conclusion in the revised manuscript to address this concern.

In page 5-6, lines 144-148, we now state:

“Noticeably, 26.89% of primary associations from our study (4,453 out of 16,562) were novel and had not been identified in prior studies (**Fig. 1G** and **Table S6**). Additionally, among these 2,339 independent pQTL regions, 2,052 regions (87.73%) have been corroborated by previously published research^[6-8], whereas 287 regions (12.27%) were newly identified, which highlights the reliability of our findings and the significance as a complement to existing resources.”

4. Line 153: “we generated a set of controls (non-pQTLs) for comparison, ensuring matching distributions of minor allele frequency to pQTLs.” No detail on how this set of non-pQTLs was generated is provided.

Response: Thanks for your valuable suggestion. The non-pQTLs dataset was randomly sampled by vSampler based on 1000 Genomes Project phase3 (1000G P3), and non-pQTLs were matched with pQTLs based on number of variants in linkage disequilibrium (LD), minor allele frequency (MAF) and variant type (**Fig.S3**). This criterion has been widely adapted in many previous published researches (Takata A et al. *Nat Commun.* 2017;8:14519; Lin D et al. *Genome Med.* 2018;10(1):13; Wang Z et al. *Genomics Proteomics Bioinformatics.* 2023;21(3):649-661) The generation of non-pQTL variants of all disease types was done in the same way.

In page 22, lines 709-713, we now state:

“For comparative analysis, we generated matched non-pQTLs for each disease type by randomly selecting genomic variants using vSampler (<http://www.mulinlab.org/vsampler/>) based on the 1000 Genomes Project Phase 3 (1000G P3). These non-pQTLs were matched to the pQTLs based on the number of linkage disequilibrium (LD)-associated variants, minor allele frequency (MAF), and variant type (**Fig. S3**).”

5. Line 173: “we employed a tiered system that classified pQTL associations into protein-specific, pathway-specific and unspecific categories.” How were the pQTL associations classified into these categories?

Response: Thank you for your attention to the pQTL classification methodology. This approach was initially detailed in the Supplementary Methods section and has now been incorporated into the Methods section in the revised manuscript:

In page 22, lines 729-734:

“Independent pQTL associations were classified following Pietzner et al^[7]. For each protein, Gene Ontology (GO) biological process terms were obtained from UniProt using all corresponding UniProt accessions. pQTLs were classified into three categories: (1) Specific-protein: associated with only one protein; (2) Specific-pathway: all associated proteins share a common GO term; (3) Non-specific: either no GO term encompasses all associated proteins, or at least one protein lacks GO annotation. This categorization was conducted independently for each disease context.”

6. Line 214: Most of the results are presented as average e.g. “colocalization analysis identified an average of six proteins”. It would be good to provide range instead of average as the it could be misleading here as the number varies a lot.

Response: We appreciate this suggestion. As requested, the revised manuscript now reports the

observed range rather than average values to address potential misinterpretation arising from data variability.

In page 7, lines 216-220, we now state:

“This identified an average of 5, proteins per disease type sharing genetic determinants (posterior probability > 0.7), ranging from 1 (e.g., uterus cancer (Uterus Ca), ovary cancer (Ovary Ca), abdominal aortic aneurysm (AAA), liver, pernicious anaemia (PA), rheumatoid arthritis (RA)) to 20 (Asthma). These findings demonstrate substantial co-localization between cis-pQTLs and disease loci (Fig. 3E and Table S12).”

7. Line 228: Term “protein coding genes (pGenes)” has been used at several occasions. It is not mention what does it mean.

Response: We apologize for the lack of clarity in our previous description. In the revised manuscript, we have now included a clearer definition of the term "pGenes" (pQTL-regulated protein-coding genes).

In page 8, lines 231-232, we now state:

"To further elucidate the functional roles of pQTL-regulated protein coding genes (pGenes) in human diseases, we performed pathway analysis on these genes."

8. Line 237: “Collectively, 98,942 survival-related pQTLs were identified”. Are these independent pQTLs? All results should be provided for independent pQTLs to be more meaningful.

Response: As per the suggestion, we now report 969 independent survival-related pQTLs identified across diseases. These results are presented in Table S13, with corresponding revisions in revised manuscript and have been updated in the corresponding public database.

Disease abbreviation	Category	Number of samples	Survival-pQTLs	Survival-pQTL pairs
Breast Ca	Cancer	8,546	39	46
Skin Ca	Cancer	28,173	28	30
Melanoma	Cancer	2,608	9	9
Colorectal Ca	Cancer	4,848	10	10
Prostate Ca	Cancer	9,850	20	20
Lymphoma	Cancer	1,836	5	5
Uterus Ca	Cancer	1,333	4	4
Head and neck Ca	Cancer	1,016	1	1
DOLM	Cardiometabolic	40,559	83	146
MACE	Cardiometabolic	31,240	67	104
AF	Cardiometabolic	22,797	33	34
Hypertension	Cardiometabolic	52,996	152	190
CHD	Cardiometabolic	28,768	81	102
T2D	Cardiometabolic	18,781	52	54
PAD	Cardiometabolic	12,723	26	29
HF	Cardiometabolic	9,766	15	16
VT	Cardiometabolic	5,309	35	42
AAA	Cardiometabolic	3,018	1	1
CS	Cardiometabolic	6,974	3	3
HS	Cardiometabolic	2,255	3	3
GERD	Digestive	35,714	12	15
Liver	Digestive	15,786	32	40
UC	Immune	1,989	1	1
RA	Immune	5,041	16	16
SLE	Immune	238	9	9
Crohn	Immune	1,049	4	5
MS	Immune	447	10	16
Osteoporosis	Musculoskeletal	12,693	17	17
Parkinson	Psychiatric_neurolog	2,266	4	4
Dementia	Psychiatric_neurolog	5,845	1	1
Asthma	Respiratory	11,984	82	113
COPD	Respiratory	16,471	66	69
Renal	Urinary	30,612	48	49

In page 8, lines 244-245, we now state:

"Collectively, 969 independent survival-related pQTLs were identified (**Table S13**)."

9. Line 392: The C-indices for models with PRS and without PRS are very similar for most of the diseases (Fig 6B), indicating that addition of PRS to the ProRS doesn't add to the prediction. Thus the conclusion that integrating genetic and proteomic data refines predictive accuracy is not correct.

Response: Thank you for this insightful comment. Although adding PRS to ProRS offers limited improvement in predictive accuracy, both PRS and ProRS, when incorporated into the basic model, lead to varying degrees of enhancement in model performance, consistent with previous reports (Helgason H et al. *JAMA*. 2023;330(8):725-735). This supports the significant potential of each in disease prediction. Accordingly, we have refined our conclusion in the revised manuscript to more accurately reflect these observations.

In page 12-13, lines 423-425, we now state:

"These findings underscore the substantial potential of genetic and proteomic data in disease prediction and highlight the value of integrated models for precise risk assessment and management strategies."

Point-by-point responses to reviewer comments

Reviewer #1 (Remarks to the Author):

Thank you for the careful revisions. I have no further comment.

Response: We thank the reviewer for the positive feedback.

Reviewer #2 (Remarks to the Author):

I commend the authors for thoroughly addressing the many comments and suggestions from the first round of revisions. I believe the manuscript is greatly improved and the analyses much more complete with the integration of these changes. I have a few additional comments & suggestions to address discrepancies in the manuscript or for potential clarification.

Response: Thanks for the positive comments. We have addressed the concerns in the following responses.

Major:

1. The authors include detail for how individual pQTLs are identified (through conditional analysis) but do not explain how pQTL regions are defined. This should be added to the Methods, especially given the discovery of almost 300 previously undetected pQTL regions. A supplementary table with summary info for each region may also be helpful, which could include region location, # of associated proteins, a list of the proteins, and the novelty of the region. While that information is technically accessible through Table S5, it is difficult to get a clear picture from that table alone which regions are newly identified.

Response: Thank you for your helpful comment. As suggested, we have added a detailed description of how pQTL regions were defined to the “Definition of pQTL regions” section in the Methods. This approach follows previous studies (Pietzner M et al. *Science*. 2021;374(6569); Sun BB et al. *Nature*. 2018;558(7708):73-79). Due to the complex and extensive linkage disequilibrium (LD) patterns, we now consistently treat the MHC region (chr6: 25.5-34.0 Mb) as a single locus, resulting in a total of 2,224 pQTL regions. All corresponding updates have been incorporated into the revised manuscript.

Additionally, to help readers locate and review regions more easily, we have sorted Table S5 (Supplementary Data4) by region number. The table already documents how each primary association compares with previous pQTL studies. To allow for direct assessment of regional novelty, we have added a new column titled “Type_region”. A region is classified as “known” if any independent pQTL within it was previously reported. This enables readers to quickly view the location, associated proteins, and novelty status of each region.

In page 20, lines 669-680, we now state:

“We defined a protein quantitative trait locus (pQTL) as a genetic locus significantly associated with protein abundance ($P < 1.72 \times 10^{-11}$). For each protein within every disease type, we: (1) collected all significant genetic variants and expanded region of ± 500 kb around each variant, (2) iteratively merged overlapping regions until no overlap remained, and (3) designated the most significant

variant in each resulting region as the sentinel variant. This window-merge procedure was implemented using BEDtools v.2.27.0. In addition, we treated the extended major histocompatibility complex (MHC) region (chr6: 25.5-34.0Mb) as a single region.

To determine whether such a region were associated with multiple plasma proteins, we adopted the “LD-based clumping approach” by Sun et al⁶. Regional sentinel variants in high LD ($r^2 > 0.8$) with each other were combined together into a single region. Finally, we obtained 2,224 independent regions across 43 disease types (**Supplementary Data4**).”

In page 5, lines 121-124, we now state:

Furthermore, to discover the independent signals associated with proteins, we performed conditional analyses with GCTA-COJO¹³, yielding a total of 25,987 independent pQTL associations across 2,224 independent genetic regions, spanning from 4 regions in leukemia to 2,130 regions in hypertension (**Fig. 1B, Supplementary Data3-4 and Supplementary Fig. 2A**).

2. Line 234-237 & Fig. 3G: Is it reasonable to suggest that these pathways (like the “Immunoregulatory interactions between a Lymphoid and a non-lymphoid cell” pathway) are contributing to multiple diseases when the pathway is also significantly enriched for proteins with pQTLs in the healthy population? It seems to me that those proteins are simply more likely to have pQTLs regardless of population characteristics, not because of a relationship to disease status.

Response: Thanks for the insightful suggestion. To clarify, we did not claim that the pathway directly causes multiple diseases. Immune pathways are fundamental components of normal physiology and are enriched for pQTLs in both healthy and disease cohorts. Accordingly, our findings point to core immune pathways as shared hubs of genetic perturbation, rather than evidence for disease causality. Genetic variants (pQTLs) within these tightly regulated networks may modulate susceptibility across diverse contexts. We have revised this sentence to prevent any potential misunderstanding.

In page 8, lines 225-229, we now state:

“Among the significantly enriched pathways, “Immunoregulatory interactions between a Lymphoid and a non-Lymphoid cell” recurred across nearly all disease types. This recurrence indicates that core immune pathways are common targets of genetic variation. Genetic perturbations in these networks appear to shape shared susceptibility patterns across diseases.”

3.Line 360: The authors give the example of a T2D-derived pQTL that is associated with T2D risk based on MR. Given this pQTL is only significant in people who already have T2D, how could it contribute to risk of developing T2D? Also notable is that AMY2B, based on Cox regression (Table S18), is associated with lower risk of T2D (HR = 0.75), which would seem to go against the MR result.

Response: Thank you for raising these important points. We addressed each point below.

(1) Why disease-biased pQTL can relate to disease risk

Genetic variants are present prior to disease onset; but the detectability of their protein effects can

be state-dependent (Farh KK, et al. *Nature*. 2015 ;518(7539):337-343). In disease-specific contexts, regulatory elements may become active—via changes in chromatin accessibility (Calderon D, et al. *Nat Genet*. 2019;51(10):1494-1505) or epigenetic modifications (Frye M, et al. *Science*. 2018;361(6409):1346-1349), thereby unmasking variant effects on protein abundance. Thus, a pQTL that is not genome-wide significant in the healthy cohort can still index a causal pathway for susceptibility.

(2) Example supporting this interpretation

AMY2B had a high missingness rate in the healthy cohort and failed quality control (QC), so no healthy-cohort pQTL could be computed, making cross-cohort comparisons uninformative. Consequently, AMY2B is unsuitable as a disease-biased pQTL example for MR. To ensure rigor, we removed proteins and SNPs that failed QC in the healthy cohort, including AMY2B, from the disease-biased pQTL analysis, thus retaining 8,278 disease-biased pQTLs.

Among QC-passed disease-biased pQTLs, BIN2 (associated with hypertension; OR = 1.29, 95% CI = 1.14-1.46) provides a more appropriate illustration. BIN2 regulates platelet calcium signaling via STIM1-IP3R axis and modulates thrombo-inflammatory response that drive vascular dysfunction in hypertension (Volz J et al. *J Clin Invest*. 2020;130(11):6064-6079). This case is consistent with the MR result and offers mechanistic insight into how disease-biased pQTL can relate to disease susceptibility. Additionally, Table S18 has been moved from the supplementary information to the pQTL database. It is now located within the “Risk Protein by cohort study” section alongside results from other models to facilitate cross-referencing.

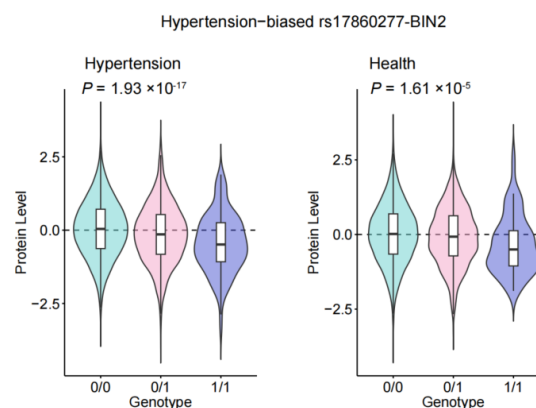


Fig. R1. Examples of disease-biased pQTL associations. The center line of the box presents as the median, the box limits indicate upper and lower quartiles and whiskers indicate the maximum and minimum.

In page 11, lines 348-351, we now state:

For example, bridging integrator 2 (BIN2) is associated with an increased risk of hypertension (OR = 1.29, 95% CI = 1.14-1.46), a finding that aligns with its function in regulating platelet calcium signaling via the STIM1-IP3R axis and modulating a thrombo-inflammatory response which contributes to vascular dysfunction¹⁹.

Minor:

1. There are a number of typos and a few incomplete sentences throughout the manuscript (e.g., line

263-264, line 421-422) that warrant correction.

Response: We thank the reviewer for highlighting these oversights. We have carefully reviewed the entire manuscript and corrected all the typos and incomplete sentences we found. Specifically, the examples on lines 263-264 and 421-422 have been rewritten for clarity and grammatical correctness.

In page 8 lines 252-254, we now state:

“We observed that the number of disease-related pQTLs increased with sample size, indicating that expanding the cohort would likely identify more associations.”

In page 13, lines 409-412, we now state:

“The ProRS model demonstrated superior performance to the PRS. Furthermore, the combined PRS+ProRS model achieved comparable or slightly better performance than the ProRS alone, with an improvement of 0.17 observed in T2D (**Fig. 6B** and **Supplementary Data18**).”

2. Given Table S4 is referenced when discussing the # of independent/primary pQTLs (line 135), it would be beneficial to add columns to Table S4 for # of primary pQTLs per disease and # of independent pQTLs for disease.

Response: Thank you for your suggestion. We have now added two new columns to **Table S4 (Supplementary Data 3):** "Number of independent pQTL associations" and "Number of primary pQTL associations," as recommended.

Supplementary Data3 : Data Summary of pQTLs for 42 disease types and healthy population in pQTL-Atlas.														
Disease abbreviation	Category	Number of samples	Number of proteins	Number of genotypes	Pairs	All pGenes	pQTLs	Pairs	cis pGenes	pQTLs	Pairs	trans pGenes	pQTLs	Number of independent pQTL
Breast Ca	Cancer	778	2,902	6,147,356	77,643	605	68,518	61,156	533	58,464	16,487	113	11,949	720
Skin Ca	Cancer	946	2,901	6,155,506	98,365	679	85,866	77,836	599	73,354	20,529	140	14,866	835
Melanoma	Cancer	178	2,902	6,115,227	11,750	175	11,460	8,898	161	8,797	2,852	16	2,733	176
Colorectal Ca	Cancer	230	2,900	6,180,116	15,130	212	14,801	12,395	196	12,204	2,735	20	2,680	213
Lung Ca	Cancer	58	2,895	5,987,734	30	5	30	29	4	29	1	1	1	5
Prostate Ca	Cancer	336	2,902	6,132,820	27,736	334	25,999	22,897	303	22,409	4,839	40	3,758	347
Lymphoma	Cancer	95	2,899	6,041,383	1,683	40	1,683	1,598	39	1,598	85	1	85	40
Uterus Ca	Cancer	94	2,898	6,014,679	1,855	53	1,855	1,766	51	1,766	89	2	89	53
Head and neck Ca	Cancer	73	2,902	6,010,390	351	17	351	350	16	350	1	1	1	17
Ovary Ca	Cancer	60	2,896	6,287,922	167	12	167	167	12	167	0	0	0	12
Leukemia	Cancer	55	2,898	5,930,514	43	4	43	40	3	40	3	1	3	4
DOLM	Cardiometabolic	6,867	2,902	6,072,663	617,319	1,551	370,101	355,241	1,297	307,322	262,078	866	100,525	4,300
MACE	Cardiometabolic	2,123	2,902	6,149,985	208,265	983	159,798	147,289	841	136,358	60,976	302	33,084	1,573
AF	Cardiometabolic	934	2,901	6,148,410	86,357	623	77,174	68,945	560	65,943	17,412	118	12,721	768
Hypertension	Cardiometabolic	12,169	2,902	6,856,353	933,939	1,822	488,283	467,844	1,485	390,740	466,095	1,249	153,053	6,994
CHD	Cardiometabolic	2,605	2,902	6,144,174	251,246	1,101	185,585	171,858	936	157,325	79,388	377	40,695	1,887
T2D	Cardiometabolic	2,407	2,902	5,966,118	200,518	1,016	153,809	141,712	878	130,466	58,806	313	32,764	1,644
PAD	Cardiometabolic	957	2,901	6,150,060	83,653	630	73,003	65,369	555	62,440	18,284	128	12,387	768
HF	Cardiometabolic	390	2,902	6,152,553	26,404	302	25,133	21,584	273	21,265	4,820	37	3,969	315
VT	Cardiometabolic	1,147	2,901	6,162,982	111,239	722	94,623	84,800	637	80,376	26,439	163	17,318	932
AAA	Cardiometabolic	61	2,901	6,091,483	150	10	150	150	10	150	0	0	0	10
CS	Cardiometabolic	202	2,902	6,147,121	11,180	156	10,963	8,534	148	8,413	2,646	10	2,604	156
HS	Cardiometabolic	130	2,901	6,157,350	7,390	82	7,351	5,201	77	5,201	2,189	6	2,189	82
GERD	Digestive	3,302	2,901	6,143,277	339,467	1,231	235,870	220,512	1,041	198,766	118,955	501	55,681	2,371
Liver	Digestive	820	2,902	6,138,545	68,145	560	61,891	53,450	493	51,563	14,695	101	11,531	646
UC	Immune	379	2,901	6,127,708	30,581	345	29,218	25,735	313	25,033	4,846	43	4,342	357
RA	Immune	649	2,902	6,155,616	54,936	458	49,732	43,161	413	42,004	11,775	70	9,069	518
SLE	Immune	370	2,900	6,121,595	23,581	290	22,563	18,575	259	18,349	5,006	38	4,356	297
Crohn	Immune	211	2,901	6,112,940	13,359	171	13,138	10,944	159	10,788	2,415	14	2,394	171
PA	Immune	190	2,902	6,128,918	12,050	167	11,779	9,616	153	9,490	2,434	16	2,340	168
MS	Immune	325	2,901	6,136,904	29,507	304	27,517	22,850	278	22,439	6,657	33	5,538	315
Osteoporosis	Musculoskeletal	1,082	2,901	6,147,227	110,866	713	93,671	83,377	624	78,619	27,489	160	18,656	905
Cataracts	Ophthalmic	1,331	2,902	6,090,445	121,324	783	103,188	94,381	687	88,904	26,843	185	18,142	1,035
Glaucoma	Ophthalmic	687	2,902	6,148,161	64,165	543	57,910	52,412	481	50,464	11,753	93	8,232	567
Depression	Psychiatric, neurologic	3,712	2,902	6,143,839	374,568	1,291	251,347	236,255	1,090	211,676	138,313	532	62,008	2,588
Parkinson	Psychiatric, neurologic	132	2,900	6,075,217	6,834	96	6,773	4,792	92	4,762	2,042	5	2,042	96
Dementia	Psychiatric, neurologic	76	2,901	6,051,533	500	27	500	499	26	499	1	1	1	27
Asthma	Respiratory	5,192	2,902	6,097,281	488,462	1,425	310,254	294,432	1,183	260,081	194,030	701	79,004	3,470
Rhinitis	Respiratory	3,908	2,902	6,103,420	389,777	1,313	264,046	247,683	1,097	221,236	142,094	592	66,191	2,787
COPD	Respiratory	1,778	2,902	6,152,982	179,127	932	139,802	127,072	804	118,360	52,055	259	28,563	1,372
Renal	Urinary	1,416	2,901	6,139,260	122,359	777	102,443	92,554	675	86,808	29,805	193	19,888	858
Gout	Urinary	961	2,901	6,157,825	91,265	666	80,314	72,569	585	69,015	18,696	132	13,451	809
Healthy population	Healthy	7,626	2,901	6,035,913	713,761	1,633	405,199	390,013	1,346	332,569	323,748	1,021	114,921	5,217

Supplementary Data 3. Data Summary of pQTLs for 42 disease types and healthy population in pQTL-Atlas.

3. Line 302: Describing this pQTL region by only rs17321515 is misleading, as searching that variant in Table S5 returns only a single pQTL instead of six like expected. Perhaps referring to the region ID (e.g., chr19_153) would make it easier for the reader to find details.

Response: Thank you for your valuable comment. To clarify, the term “Trans-pQTL hotspots” in the original manuscript refers to specific genetic variant sites, not broader genomic regions, which is why we used the rs17321515 identifier. And these hotspots are derived from the overall pQTL findings rather than independent results. For completeness, all significant pQTL results associated

with rs17321515 are provided in **Table R1**.

SNP ID	Proteins	Pvalue	Beta	pQTL types
rs17321515	IL1R2	7.82E-21	-0.117460729	trans
rs17321515	CHI3L1	1.89E-17	-0.103131869	trans
rs17321515	CEACAM16	6.05E-15	0.096598972	trans
rs17321515	FETUB	1.81E-14	-0.09239253	trans
rs17321515	CA14	5.82E-12	0.085230878	trans
rs17321515	CPM	1.17E-11	-0.082628242	trans

Table R1. Trans-pQTL results associated with rs17321515 ($P < 1.72 \times 10^{-11}$)

Additionally, following your suggestion, we have now defined the trans-pQTL hotspots by genomic region (e.g., chr8_77), which should make it easier for readers to locate the corresponding data (Supplementary Data4). The region_ID chr8_77 is used to refer to the trans hotspot region mentioned above in revised manuscript.

In page 9-10, lines 289-298, we now state:

“Notably, we identified trans-hotspot regions that were associated with multiple proteins yet confined to a single disease cluster, underscoring their shared genetic basis. For example, region chr8_77 is associated with nine proteins in trans, and crucially, these proteins are associated with either hypertension or DOLM, both mapping to cluster 6. (**Supplementary Fig. 4H**). By acting through multiple proteins in a specific disease context, this hotspot amplifies the co-morbidity pathological network, particularly in processes such as protein homeostasis, amino acid transport, and neuroendocrine and inflammatory signaling. This reveals an interconnected network of cardiovascular, metabolic, and immune dysregulation (**Fig. 4E**), offering crucial insights for understanding the genetic basis of these conditions and for identifying new therapeutic targets.”

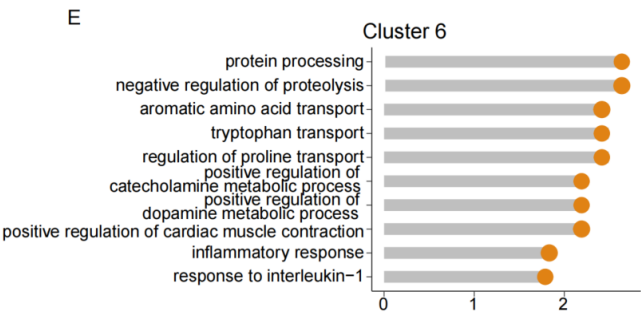


Fig.4E The results of Gene Ontology (GO) biological process (BP) enrichment proteins associated with the trans-pQTL hotspot region chr8_77.

4. Line 355: Is the 68.37% number representing the proportion of healthy results that are significant in MR analyses using disease-derived pQTLs? This should be clarified.

Response: Thank you for your comment. Because instrumental variables (IVs) can be derived from pQTLs in both healthy and disease cohorts, we performed MR analyses using each set of IVs separately. In total, 201 protein-disease associations were significant under both instrument frameworks, representing 68.37% (201/294) of the MR findings based on disease-derived pQTLs. We clarified this with a Venn diagram.

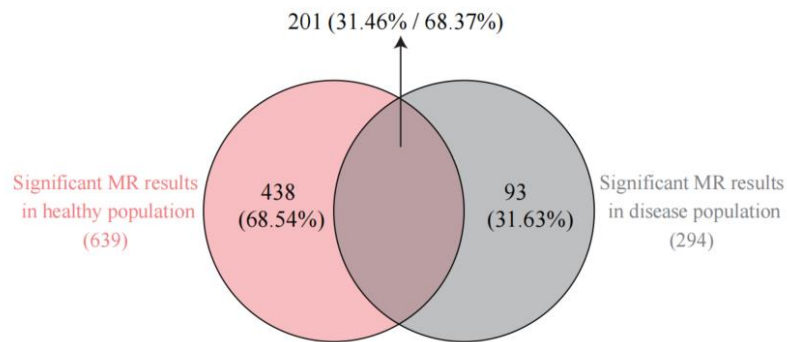


Fig.S7 The Venn diagram compares the Mendelian randomization (MR) results between the healthy and disease cohorts. The left pink circle represents the number of significant protein-disease associations identified in the healthy cohort, while the right gray circle shows those from the disease cohort. The overlapping section indicates protein-disease associations identified in both cohorts.

In page 11, lines 343-346, we now state:

“Furthermore, 68.37% (201/294) of MR associations instrumented by disease-derived pQTLs were also significant when instrumented by healthy-derived pQTLs, thereby reinforcing and expanding the MR results (**Supplementary Data14, Supplementary Data15 and Supplementary Fig. 7**).”

5. Line 462: Are the numbers presented here flipped, i.e. it should be 8,380 disease-related and 1,806 disease-specific pQTLs?

Response: Thank you for your comment. The numbers are not flipped. We identified a total of 1,806 disease-related pQTL associations, as well as 8,380 disease-biased pQTL associations. This is consistent with the description in the result section. In addition, in response to major Comment 3, we further refined the disease-biased pQTL analysis by removing proteins and SNPs that failed QC in the healthy cohort. This rigorous approach yielded a final set of 8,278 pQTL associations, and corresponding revisions have been incorporated into the manuscript.

In page 14, lines 450-452, we now state:

This analysis identified 1,806 disease-related pQTL associations and 8,278 associations biased to particular diseases.

6. Line 861 & 866: The references to these tables are no longer correct.

Response: Thank you for your careful comment. I sincerely apologize again for our carelessness. We replaced the correct table numbers.

In page 24, lines 847-851, we now state:

“To assess potential pleiotropy impact, we scrutinized it further using the MR-Egger and MR-PRESSO methods. Additionally, we implemented other prevalent MR methods (weighted median, weighted mode, and simple mode) to enhance the robustness of our findings (**Supplementary Data14, Supplementary Data15**).”

In page 25, lines 854-857, we now state:

“To evaluate the robustness of these associations, we conducted additional checks by integrating the results from protein-disease associations with those obtained from MR. Proteins consistently showing associations in both analyses were retained as “high-confidence proteins” (Supplementary Data16).”

7. Figure 2A: In the enrichment portion of the plot, some cells (i.e., synonymous variants for breast cancer) are significantly enriched for pQTLs, yet the color suggests a $\log_2(\text{OR})$ of 0. While this is likely because there are no synonymous variants in the non-pQTL variants selected (leading to an infinite OR), the presentation is misleading. Perhaps the color should be changed for those examples.

Response: We thank the reviewer for pointing this out. We have fixed the misleading presentation of infinite values by converting the $\log_2(\text{OR})$ into a segmented scale that explicitly includes and visualizes both $-\text{Inf}$ and Inf values. This correction has been applied to Figures 2B, 2C, and 4C. The legend has also been modified to reflect this change.

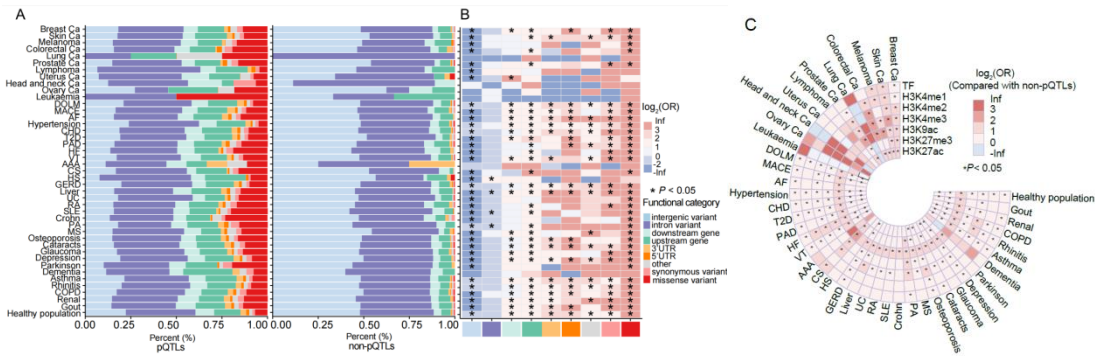


Fig.2B-C. Enrichment analysis of cis-pQTLs compared with cis non-pQTLs among variants within functional category. Results were calculated using the two-tailed Fisher’s exact test. C. or 7 epigenomic marks identified by ENCODE in each disease type. $\log_2(\text{OR}) > 0$ ($\text{OR} > 1$) indicating enrichment and < 0 ($\text{OR} < 1$) not-enrichment. Categories significantly enriched for cis-pQTLs are indicated with an asterisk (two-tailed Fisher’s-exact test).

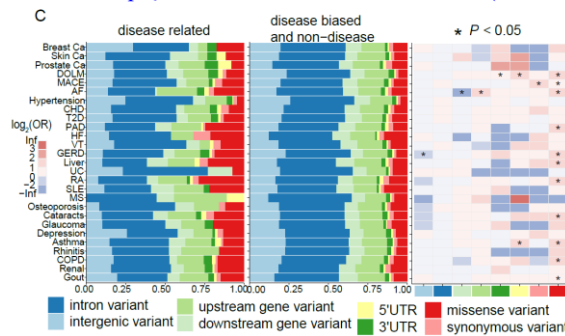


Fig4C. Genomic distribution of disease-specific (left), disease biased and non-disease (middle) pQTLs, alongside enrichment analysis comparing the two. Results were calculated using the two-tailed Fisher’s exact test.

8. Figure 2D: Are these plots properly ordered? In lung cancer, for example, the top plot (labeled cis-pQTL) looks more like the distribution when combining the 2nd and 3rd panels of the plot.

Response: Thank you for your thoughtful comment. The plots are correctly ordered—top: cis-pQTLs, middle: trans-pQTLs, bottom: all pQTLs. The explained variance of “all pQTLs” is estimated by a combined model including both cis and trans-pQTLs. It is typically close to the sum

of the variances explained by cis and trans-pQTLs, and therefore higher than either component alone. Consequently, its distribution is not a visual overlay of the first two panels; instead, it tends to track the dominant contributor. For lung cancer, cis is the dominant contributor. Exact confidence intervals are provided in **Supplementary Data6**.

Reviewer #3 (Remarks to the Author):

Authors have adequately addressed all the concerns. I have no more comments.

Response: We thank the reviewer for the positive evaluation of our work.